

Binus University

Presentation Final Project

PT. SUPERINDO

by Group 2

About Superindo



Superindo adalah jaringan supermarket terpercaya yang telah hadir sejak 1997, dengan lebih dari 200 gerai di berbagai kota besar di Indonesia. Superindo menyediakan produk kebutuhan sehari-hari yang segar, berkualitas, dan terjangkau.

Superindo fokus pada kenyamanan dan kepuasan pelanggan melalui layanan responsif dan lingkungan belanja yang bersih dan nyaman. Superindo juga menerapkan **proses manajemen pelanggan** yang bertujuan membangun loyalitas dan meningkatkan pengalaman belanja. Proses ini mencakup identifikasi kebutuhan pelanggan, layanan yang cepat dan ramah, program keanggotaan, serta pengumpulan feedback.

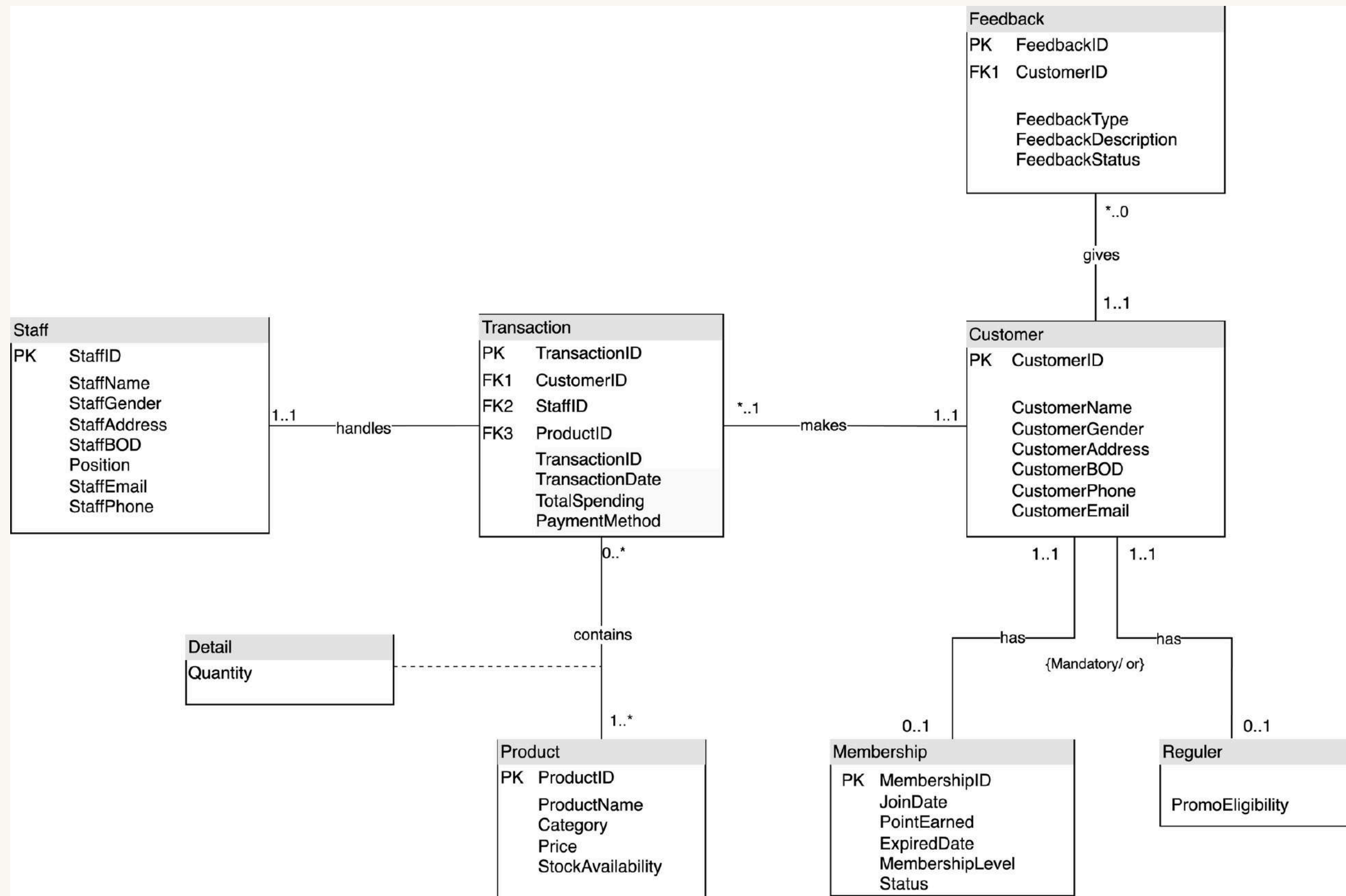
Melalui **program membership**, pelanggan mendapatkan berbagai keuntungan eksklusif dan promo menarik, sebagai bentuk apresiasi sekaligus strategi untuk memperkuat hubungan jangka panjang dengan pelanggan.

Data and Information Management

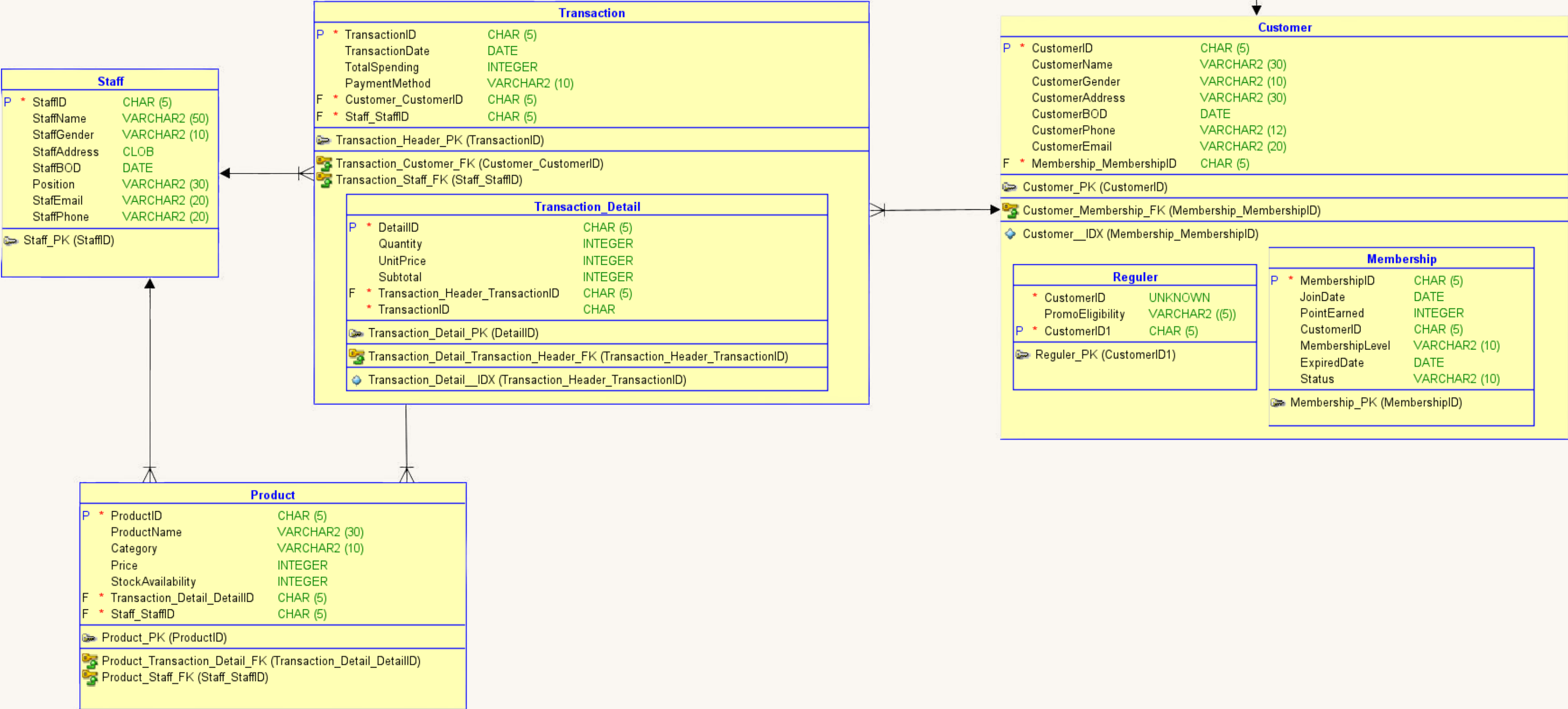
Final Logical ERD

Group 2

UML



Oracle Academy

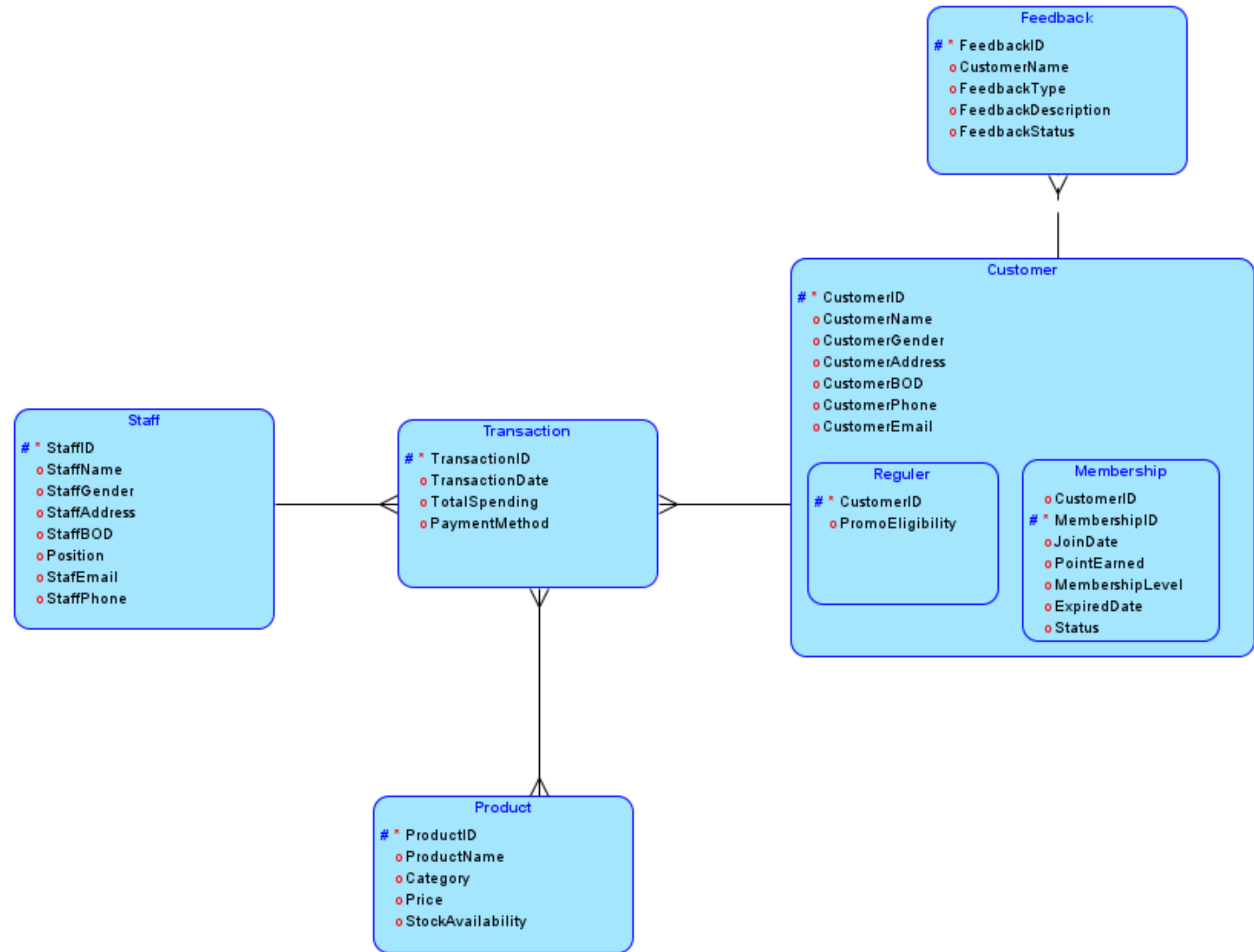


Data and Information Management

Proses Normalisasi

Group 2

Logical Database



Normalization

UNF

Transact ionID	Transactio nDate	Custo merID	StaffID	Productl D	ProductN ame	Quantit y	Price	Discoun t	Subtotal
TR001	01/06/2025	CU001	ST001	PU001, PU002	Indomie, Ultra Milk	2, 1	3500, 25000	-	7000, 25000

1NF

Transactionl D	Transaction Date	Customer ID	Staf fID	Productl D	ProductN ame	Quantity	Price	Discount	Subtotal
TR001	01/06/2025	CU001	ST0 01	PU001	Indomie	2	3500	-	7000
TR001	01/06/2025	CU001	ST0 01	PU002	Ultra Milk	1	25000	-	25000

Normalization

2NF

Transaction Header

TransactionID	TransactionDate	CustomerID	StaffID
TR001	01/06/2025	CU001	ST001

Transaction Detail

DetailID	TransactionID	ProductID	Quantity	Price	Discount	Subtotal
TD001	TR001	PU001	2	3500	-	7000
TD002	TR001	PU002	1	25000	-	25000

3NF

ProductID	ProductName	Category	Price	Discount	StockAvailability
PU001	Indomie	10	3500	-	500
PU002	Ultra Milk	11	25000	-	100

Data and Information Management

Physical Database

Group 2

Design Base Relations

Domain CustomerID: number
Domain CustomerName: variable length character string, length 50
Domain CustomerGender: variable length character string, length 10
Domain CustomerPhone: variable length character string, length 12
Domain CustomerBOD: date
Domain CustomerEmail: variable length character string, length 20

Domain StaffID: number
Domain StaffName: variable length character string, length 50
Domain StaffGender: variable length character string, length 10
Domain StaffAddress: character large object (CLOB)
Domain StaffDOB: date
Domain Position: variable length character string, length 30
Domain StaffEmail: variable length character string, length 20
Domain StaffPhone: variable length character string, length 20

Domain ProductID: number
Domain ProductName: variable length character string, length 50
Domain Category: variable length character string, length 10
Domain Price: integer
Domain StockAvailability: integer

Domain TransactionID: number
Domain TransactionDate: date
Domain TotalSpending: integer
Domain PaymentMethod: variable length character string, length 10

Design Base Relations

Customer (CustomerID, CustomerName, CustomerGender, CustomerPhone, CustomerBOD, CustomerEmail)

```
CREATE TABLE customer (  
  CustomerID    NUMBER NOT NULL,  
  CustomerName  VARCHAR2(50) NOT NULL,  
  CustomerGender VARCHAR2(10),  
  CustomerPhone VARCHAR2(12),  
  CustomerBod   DATE,  
  CustomerEmail VARCHAR2(20),  
  PRIMARY KEY (CustomerID)  
);
```

Staff (StaffID, StaffName, StaffGender, StaffAddress, StaffDOB, Position, StaffEmail, StaffPhone)

```
CREATE TABLE staff (  
  StaffID      NUMBER NOT NULL,  
  StaffName    VARCHAR2(50) NOT NULL,  
  StaffGender  VARCHAR2(20),  
  StaffAddress TEXT,  
  StaffBOD     DATE,  
  Position     VARCHAR2(30),  
  StaffEmail   VARCHAR2(20),  
  StaffPhone   VARCHAR2(20),  
  PRIMARY KEY (StaffID)  
);
```


Design Base Relations

Product (ProductID, ProductName, Category, Price, StockAvailability)

```
CREATE TABLE product (  
  ProductID      NUMBER NOT NULL,  
  ProductName    VARCHAR2(30) NOT NULL,  
  Category       VARCHAR2(10),  
  Price          INTEGER NOT NULL,  
  StockAvailability INTEGER NOT NULL,  
  PRIMARY KEY (ProductID)  
);
```

Transaction (TransactionID, TransactionDate, TotalSpending, PaymentMethod)

```
CREATE TABLE transaction (  
  TransactionID   NUMBER NOT NULL,  
  TransactionDate DATE NOT NULL,  
  TotalSpending   INTEGER NOT NULL,  
  PaymentMethod   VARCHAR2(10),  
  CustomerID      NUMBER NOT NULL,  
  StaffID         NUMBER NOT NULL,  
  PRIMARY KEY (TransactionID),  
  FOREIGN KEY (CustomerID) REFERENCES customer(CustomerID),  
  FOREIGN KEY (StaffID) REFERENCES staff(DtaffID)  
);
```

Design Base Relations

Membership (MembershipID, JoinDate, PointEarned, Membership_Level, Expired_Date, Status)

```
CREATE TABLE Membership (  
  MembershipID      CHAR(5) NOT NULL,  
  JoinDate          DATE NOT NULL,  
  PointEarned       INTEGER,  
  MembershipLevel   VARCHAR2(10) NOT NULL,  
  ExpiredDate       DATE NOT NULL,  
  Status            VARCHAR2(10) NOT NULL,  
  PRIMARY KEY (MembershipID)  
);
```

Regular (CustomerID, Promo_Eligibility)

```
CREATE TABLE Regular (  
  CustomerID        CHAR(5) NOT NULL,  
  PromoEligibility  VARCHAR2(10),  
  PRIMARY KEY (CustomerID)  
);
```

Feedback (FeedbackID, FeedbackType, FeedbackDescription, FeedbackStatus)

```
CREATE TABLE Feedback (  
  FeedbackID        CHAR(5) NOT NULL,  
  FeedbackType      VARCHAR2(20),  
  FeedbackDescription CLOB,  
  FeedbackStatus    VARCHAR2(20) NOT NULL,  
  PRIMARY KEY (FeedbackID)  
);
```

Design Derive Data

Customer & Transaction: Total Spending per Customer

CustomerID	CustomerName	TotalSpending
CU001	Anna Elsa	850000
CU002	Jason Crus	675000

```
SELECT
  c.CustomerID,
  c.CustomerName,
  SUM(t.TotalSpending) AS TotalSpending
FROM
  Customer c
JOIN
  Transaction t ON c.CustomerID = t.CustomerID
GROUP BY
  c.CustomerID, c.CustomerName;
```

Design Derive Data

Customer & Membership: Point Summary

CustomerID	CustomerName	TotalPoints
CU001	Anna Elsa	120
CU002	Dina	90

```
SELECT
  c.CustomerID,
  c.CustomerName,
  SUM(m.PointEarned) AS TotalPoints
FROM
  Customer c
JOIN
  Membership m ON c.CustomerID = m.CustomerID
GROUP BY
  c.CustomerID, c.CustomerName;
```


Design Derive Data

Product & Transaction: Product Sales Count

ProductID	ProductName	TotalSold
PU001	Indomie	100
PU002	Ultra Milk	50

```
SELECT
  p.ProductID,
  p.ProductName,
  COUNT(t.TransactionID) AS TotalSold
FROM
  Product p
JOIN
  Transaction t ON p.ProductID = t.ProductID
GROUP BY
  p.ProductID, p.ProductName;
```

Design Derive Data

Customer & Feedback: Feedback Count by Type

FeedbackType	FeedbackType	FeedbackCount
FB001	Complaint	5
FB002	Suggestion	3
FB003	Praise	2

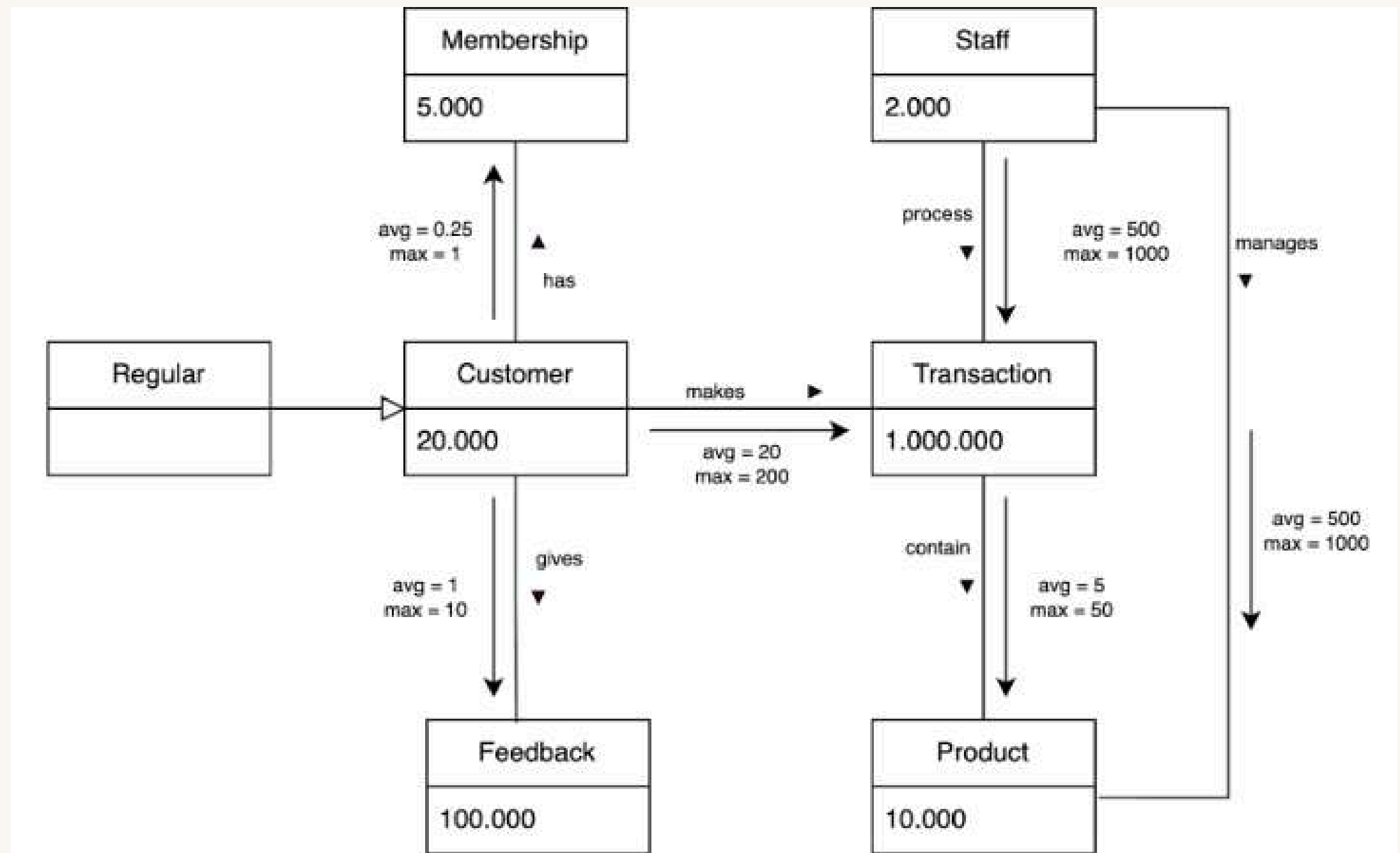
```
SELECT
    FeedbackType,
    COUNT(*) AS FeedbackCount
FROM
    Feedback
GROUP BY
    FeedbackType;
```

Analyze Transaction

Superindo merupakan ritel supermarket yang berkembang di Indonesia dan menjadi supermarket berkualitas dalam pelaksanaan transaksinya. Untuk menganalisa transaksi yang ada di Superindo dalam satu tahunnya kita bisa membuat dugaan/perkiraan rata-rata customer yang melakukan transaksi pembelian produk. Penghitungannya adalah sebagai berikut:

Menghitung setiap data pada setiap Entitas Utama:

- 1.Customer (20.000)
- 2.Staff (2.000)
- 3.Transaction (1.000.000)
- 4.Product (10.000)
- 5.Membership (5.000)
- 6.Feedback (100.000)

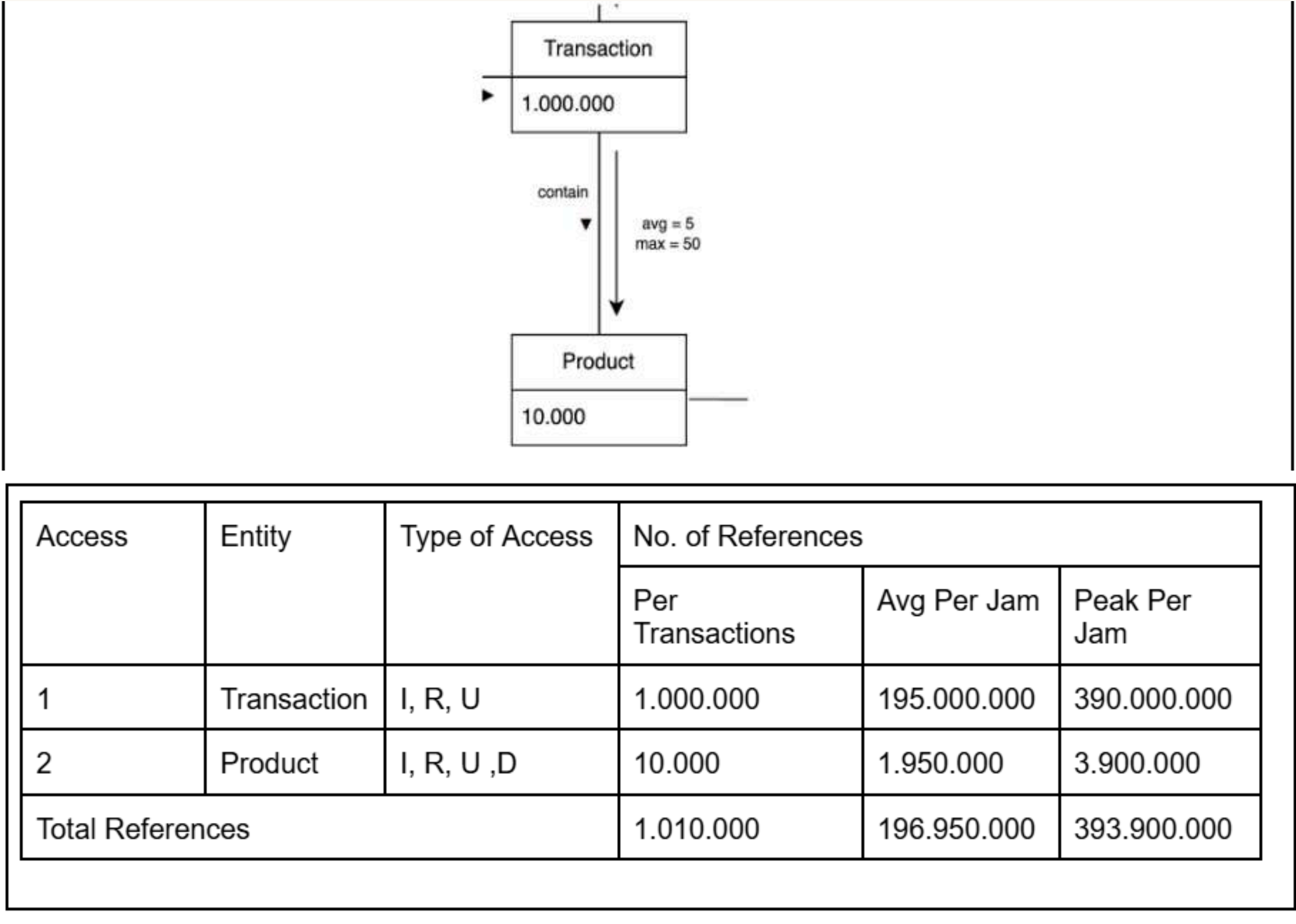


Analyze Transaction

Satu proses transaksi akan melibatkan database entiti berupa customer, transaksi, produk, dan staff serta Feedback dan Membership. Berikut Transaction Analysis Form:

Transaction Analysis Form	
Transaction (C) List ProductID, ProductName, dan jumlah pembelian dari transaksi selama 1 tahun terakhir. Urutkan berdasarkan jumlah pembelian terbanyak.	
Transaction Volume	
Average = 195 per jam	
Peak = 390 per jam (di antara jam 17.00-19.00 di awal bulan/liburan)	
<pre>SELECT p.ProductID, p.ProductName, COUNT(tp.ProductID) AS TotalPurchased FROM transaction t JOIN transaction_product tp ON t.TransactionID = tp.TransactionID JOIN product p ON tp.ProductID = p.ProductID WHERE t.TransactionDate >= ADD_MONTHS(SYSDATE, -12) GROUP BY p.ProductID, p.ProductName ORDER BY TotalPurchased DESC;</pre>	Predicate: t.TransactionDate >= ADD_MONTHS(SYSDATE, -12) Join attributes : t.TransactionID = tp.TransactionID, tp.ProductID = p.ProductID Ordering attribute: TotalPurchased Grouping attribute: p.ProductID, p.ProductName Built-in functions : none Attributes update : ProductAvailability

Analyze Transaction



Estimate Disk Space

Rincian Ukuran Kolom dari Tabel Customer

Column Name	Data Type	Bytes
CustomerID	CHAR(5)	5 bytes
CustomerName	VARCHAR(50)	50 bytes
CustomerGender	VARCHAR(10)	10 bytes
CustomerBOD	DATE	3 bytes
CustomerPhone	VARCHAR(12)	12 bytes
CustomerEmail	VARCHAR(20)	20 bytes
CustomerAddress	TEXT	100 bytes

- Total per Entri = $5 + 50 + 10 + 3 + 12 + 20 + 100 = 200$ bytes
- Total Ukuran Tabel =

Jika diasumsikan ada 20.000 entri customer,

→ Total Ukuran Tabel = $20.000 \text{ entri} \times 200 \text{ bytes/entri} = 4.000.000 \text{ bytes}$ (atau 4 MB)

- Ukuran tabel Customer diperkirakan sekitar 4 MB untuk 20.000 entri.

Estimate Disk Space

Rincian Ukuran Kolom dari Tabel Staff

Column Name	Data Type	Bytes
StaffID	CHAR(5)	5 bytes
StaffName	VARCHAR(50)	50 bytes
StaffGender	VARCHAR(10)	10 bytes
StaffAddress	TEXT	100 bytes
Position	VARCHAR(30)	30 bytes
StaffPhone	VARCHAR(20)	20 bytes
StaffEmail	VARCHAR(20)	20 bytes

- Total per Entri = $5 + 50 + 10 + 100 + 30 + 20 + 20 = 235$ bytes
- Total Ukuran Tabel =

Jika diasumsikan ada 2.000 entri staff,

→ Total Ukuran Tabel = $2.000 \text{ entri} \times 235 \text{ bytes/entri} = 470.000 \text{ bytes}$ (atau 0.47 MB)

- Ukuran tabel Staff diperkirakan sekitar 0.47 MB untuk 2.000 entri.

Estimate Disk Space

Rincian Ukuran Kolom dari Tabel Transaction

Column Name	Data Type	Bytes
TransactionID	CHAR(5)	5 bytes
StaffID	CHAR(5)	5 bytes
CustomerID	CHAR(5)	5 bytes
TransactionDate	DATE	3 bytes
TotalSpending	INTEGER	4 bytes
PaymentMethod	VARCHAR(10)	10 bytes
Quantity	INTEGER	4 bytes

- Total per Entri = $5 + 5 + 5 + 3 + 4 + 10 + 4 = 36$ bytes
- Total Ukuran Tabel =

Jika diasumsikan ada 1.000.000 entri Transaction,

→ Total Ukuran Tabel = $1.000.000 \text{ entri} \times 36 \text{ bytes/entri} = 36.000.000 \text{ bytes}$ (atau 36 MB)

- Ukuran tabel Transaction diperkirakan sekitar 36 MB untuk 1.000.000 entri.

Estimate Disk Space

Rincian Ukuran Kolom dari Tabel Product

Column Name	Data Type	Bytes
ProductID	CHAR(5)	5 bytes
ProductName	VARCHAR(30)	30 bytes
Category	VARCHAR(10)	10 bytes
Price	INTEGER	4 bytes
StockAvailability	INTEGER	4 bytes

- Total per Entri = $5 + 30 + 10 + 4 + 4 = 53$ bytes
- Total Ukuran Tabel =

Jika diasumsikan ada 10.000 entri product,

→ Total Ukuran Tabel = $10.000 \text{ entri} \times 53 \text{ bytes/entri} = 530.000 \text{ bytes}$ (atau 0.53 MB)

- Ukuran tabel Product diperkirakan sekitar 0.53 MB untuk 10.000 entri.

Estimate Disk Space

Rincian Ukuran Kolom dari Tabel Membership

Column Name	Data Type	Bytes
MembershipID	CHAR(5)	5 bytes
CustomerID	CHAR(5)	5 bytes
JoinDate	DATE	3 bytes
PointEarned	INTEGER	4 bytes
MembershipLevel	VARCHAR(10)	10 bytes
ExpiredDate	DATE	3 bytes
Status	VARCHAR(10)	10 bytes

- Total per Entri = $5 + 5 + 3 + 4 + 10 + 3 + 10 = 40$ bytes

- Total Ukuran Tabel =

Jika diasumsikan ada 5.000 entri membership,

→ Total Ukuran Tabel = $5.000 \text{ entri} \times 40 \text{ bytes/entri} = 200.000 \text{ bytes}$ (atau 0.2 MB)

- Ukuran tabel Membership diperkirakan sekitar 0.2 MB untuk 5.000 entri.

Estimate Disk Space

Rincian Ukuran Kolom dari Tabel Regular

Column Name	Data Type	Bytes
CustomerID	CHAR(5)	5 bytes
PromoEligibility	VARCHAR(10)	10 bytes

- Total per Entri = 5 + 10 + = 15 bytes
- Total Ukuran Tabel =

Jika diasumsikan ada 15.000 entri regular,

→ Total Ukuran Tabel = 15.000 entri x 15 bytes/entri = 225.000 bytes (atau 0.225 MB)

- Ukuran tabel Regular diperkirakan sekitar 0.225 MB untuk 15.000 entri.

Rincian Ukuran Kolom dari Tabel Feedback

Column Name	Data Type	Bytes
FeedbackID	CHAR(5)	5 bytes
CustomerID	CHAR(5)	5 bytes
FeedbackType	VARCHAR(20)	20 bytes
CustomerName	VARCHAR(50)	50 bytes
FeedbackDescription	TEXT	100 bytes
FeedbackStatus	VARCHAR(20)	20 bytes

- Total per Entri = 5 + 5 + 20 + 50 + 100 + 20 = 200 bytes
 - Total Ukuran Tabel =
- Jika diasumsikan ada 100.000 entri Feedback,,
→ Total Ukuran Tabel = 100.000 entri x 200 bytes/entri = 20.000.000 bytes (atau 20 MB)
- Ukuran tabel Feedback diperkirakan sekitar 20 MB untuk 100.000 entri.

Estimate Disk Space

Perhitungan size database untuk 5 tahun ke depan

Tabel	Tahun ke-1	Tahun ke-2	Tahun ke-3	Tahun ke-4	Tahun ke-5	Total 5 Tahun
Customer	4 MB	8 MB	12 MB	16 MB	20 MB	60 MB
Staff	0.46 MB	0.94 MB	1.41 MB	1.88 MB	2.35 MB	7.05 MB
Transaction	36 MB	72 MB	108 MB	144 MB	180 MB	540 MB
Product	0.53 MB	1.06 MB	1.59 MB	2.12 MB	2.65 MB	7.95 MB
Membership	0.2 MB	0.4 MB	0.6 MB	0.8 MB	1.0 MB	3.0 MB
Regular	0.225 MB	0.45 MB	0.675 MB	0.9 MB	1.115 MB	3..375 MB
Feedback	20 MB	40 MB	60 MB	80 MB	100 MB	300 MB

Total Ukuran Database:

Tahun ke-1: 61.415 MB
Tahun ke-2: 122.85 MB
Tahun ke-3: 184.275 MB
Tahun ke-4: 245.7 MB
Tahun ke-5: 307.115 MB
Total 5 Tahun: 921.375 MB
≈ 0.93 GB

Create Cross-Reference Transactions

Transaction/Relation	(A)	(B)	(C)	(D)	(E)	(F)
	I R U D	I R U D	I R U D	I R U D	I R U D	I R U D
Customer	X X	X		X X		
Staff	X X			X X		
Product	X X X X	X	X	X X X X		
Transaction	X X	X	X			
Membership	X X X			X X X	X	
Regular	X X X			X X X	X	
Feedback	X X			X X		X

I = Insert
R = Read
U = Update
D = Delete

Penjelasan

- (A) misalnya transaksi Input Data Customer Baru → insert & read ke tabel Customer, Staff, Product, Membership, dll.
- (B) misalnya transaksi Hanya Read Report → lebih ke Product, Customer, Transaction.
- (C) misalnya transaksi Update Transaksi → hanya Transaction, Product update stok.
- (D) misalnya transaksi Delete Product atau Customer.
- (E) transaksi Update Membership.
- (F) transaksi Hapus Feedback Lama.

Design User Views

- **Transaction**, consisting of the Cashier, Supervisor, Store Manager, and Finance Staff User Views;
- **Customer**, consisting of the Customer Service, Marketing, and Membership Admin User Views;
- **Staff**, consisting of the HR Manager, Supervisor, and Assistant User Views;
- **Product**, consisting of the Inventory Staff, Purchasing Staff, and Warehouse Manager User Views;
- **Membership**, consisting of the Membership Admin and Marketing User Views;
- **Feedback**, consisting of the Customer Service and Quality Assurance User Views;
- **Regular**, consisting of the Marketing and Promo Manager User Views;

Data and Information Management

User Interface & Transaction Design

Group 2

HOME PAGE



Home

Product

Cart

Feedback

Search



Jungkook



Today's Fresh
VEGETABLES

Lauk segar hemat!

13 - 18 May 2025

35% All Item!



History



Orders : #0001

Item Quantity : 5

Total Price :
Rp120.000,-

13 May 2025



Orders : #0002

Item Quantity : 8

Total Price :
Rp213.500,-

7 May 2025

Catalog



Shop by categories



FRUITS



VEGETABLES



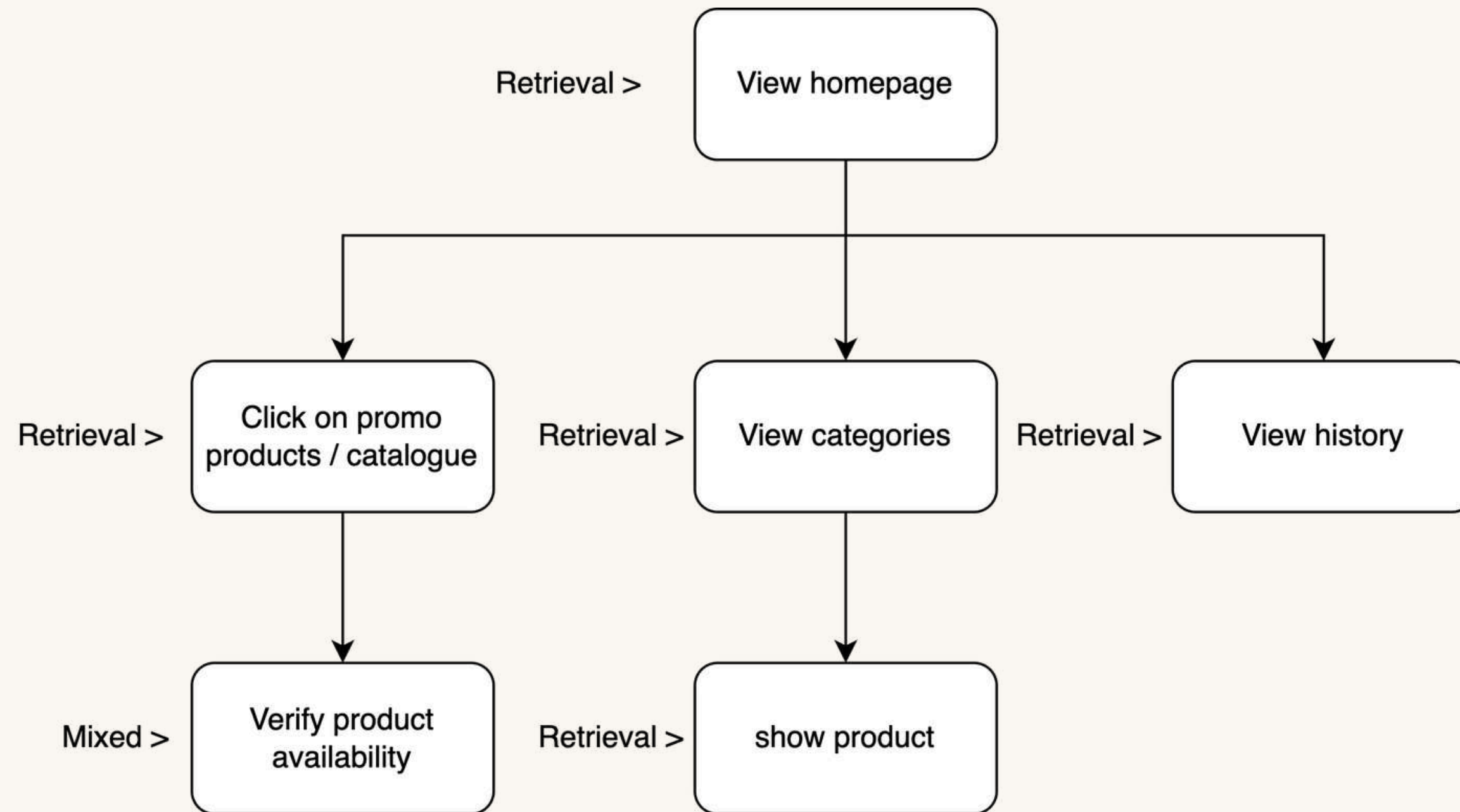
MEAT & FISH



SNACKS



Homepage Transaction Design



PROFILE

Profile

Home

Product

Cart

Feedback

Profile



Hi, Jungkook!

Silver

187 points

187 points

1000 points

Membership ID

MB197

Status Membership

Active

Join Date

5 July 2023

Expired Date

5 July 2028

Name

Jeon Jungkook

Birth of Date

1 September 1997

Gender

☒ Male

☐ Female

No Phone

081234567890

Email

Jungkook.jeon@gmail.com

Province

East Java

City

Malang

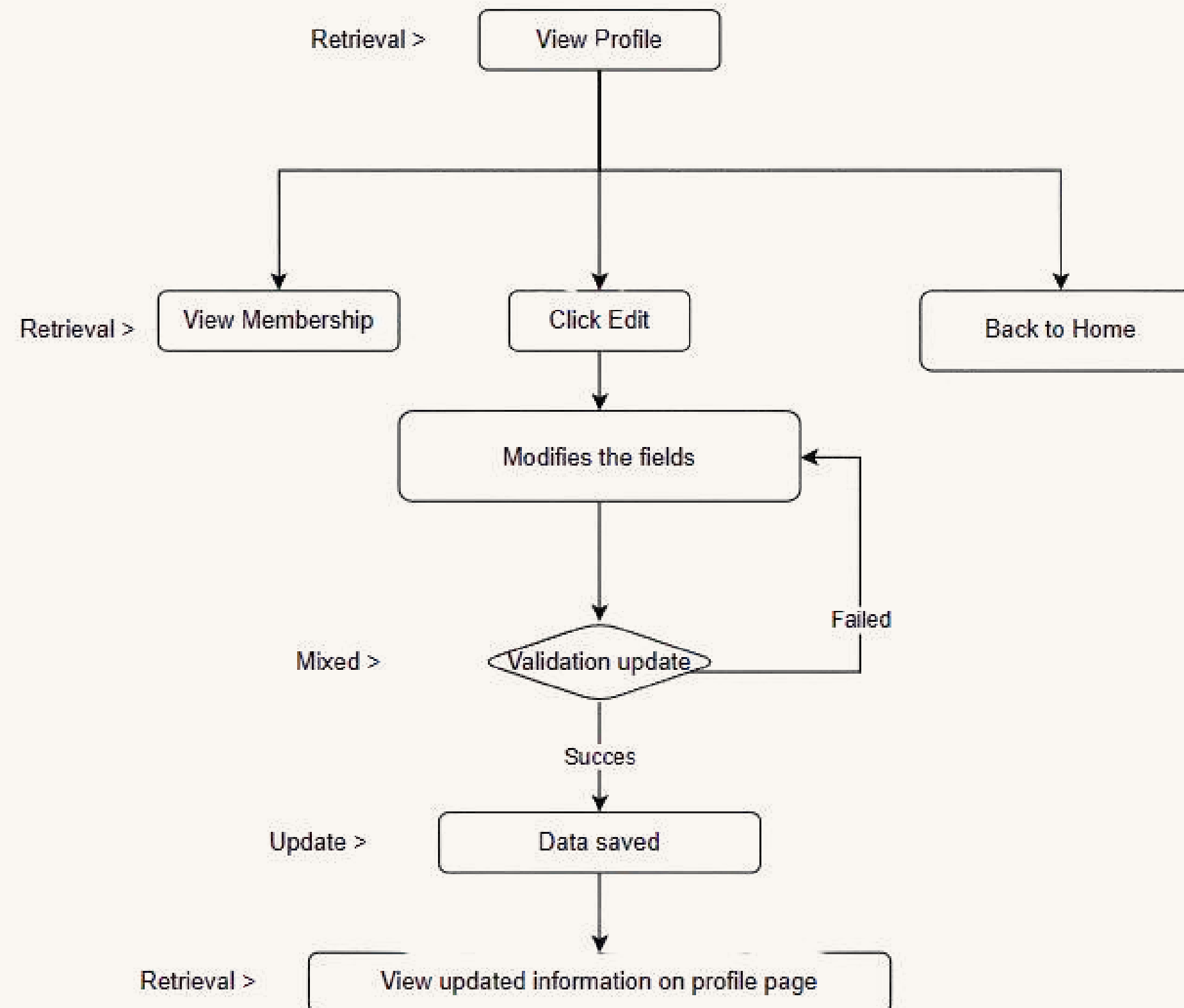
Address

Cempaka Putih street No. 143 - Blimbing, Malang, East Java, Indonesia

BACK

EDIT

Profile Transaction Design



PRODUCT

Home

Product

Cart

Feedback

Jungkook

Product

All Categories ▾

Fruits ▾

Vegetables ▾

Milk & Dairy ▾

Search product...



Naturally Fresh Avocado
1.000 gr
Rp39.900
Add to cart



Pineapple
pcs
Rp15.000
- +



Kiwi Gold
1.000 gr
Rp149.500
Add to cart



Banana
1.000 gr
Rp24.400
Add to cart



Grapes
1.000 gr
Rp79.500
Add to cart

Best Offer



Orange Valencia
1.000 gr
Rp29.800 ~~Rp44.900~~
Add to cart



Pear Golden
1.000 gr
Rp32.500
Add to cart



Watermelon
1.000 gr
Rp14.700
Add to cart

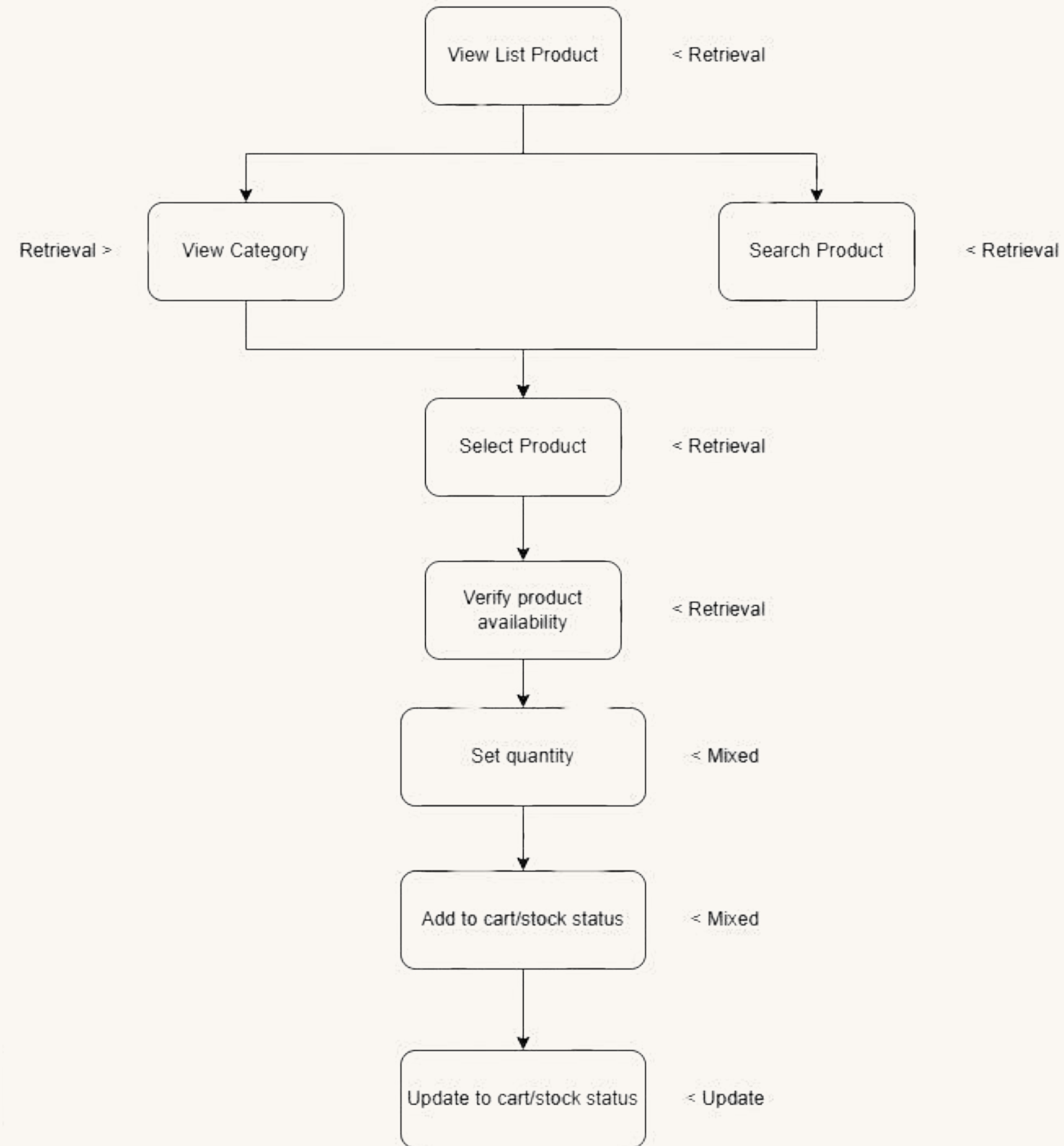








Product Transaction Design





Home

Product

Cart

Feedback

Cart



Pineapple
Rp15.000

- 1 +



Banana
Rp24.400

- 1 +



Grapes
Rp79.500

- 2 +



Naturally Fresh Avocado
Rp39.900

- 3 +



Jungkook



SUMMARY

Order total **Rp318.100**

Voucher code **None used**

Shipping **Rp15.000**

Rp333.100

We accept



VISA

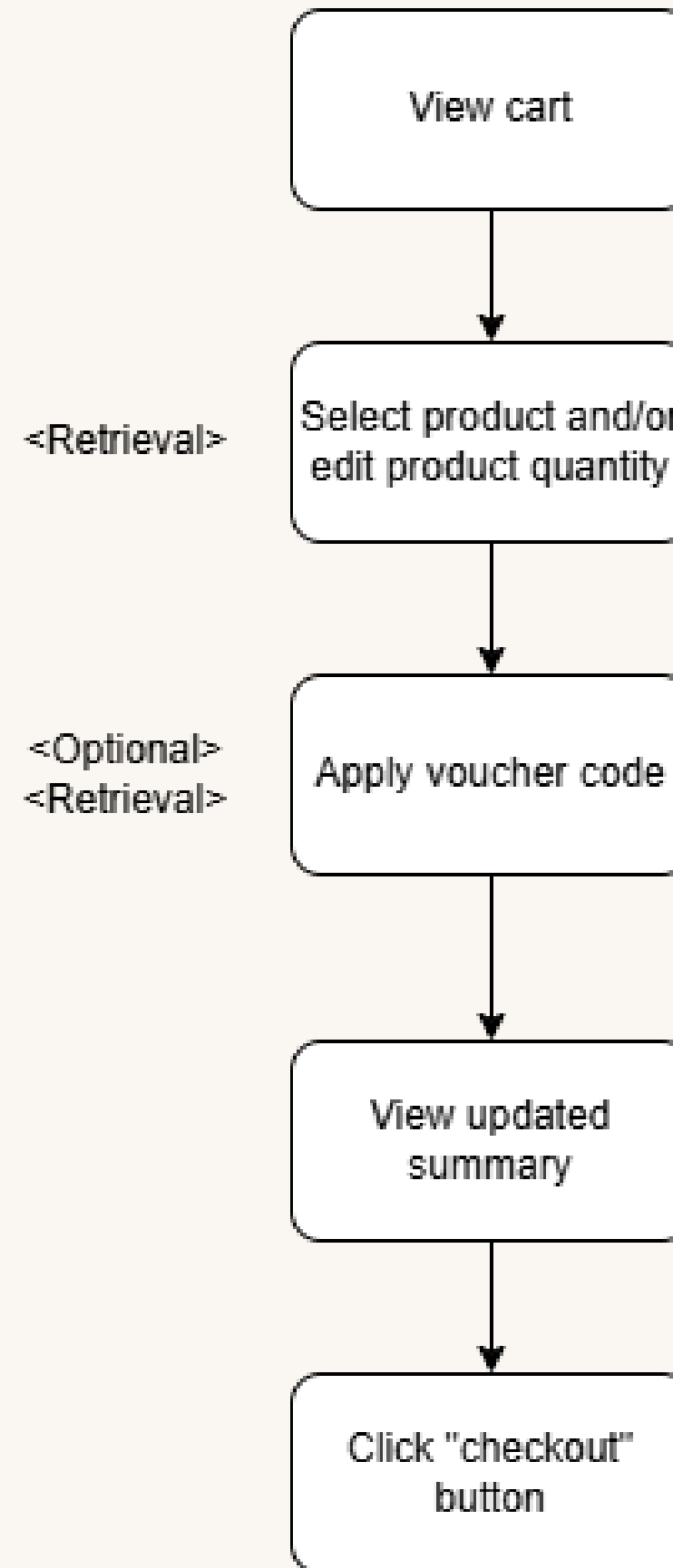


IRIS

CHECKOUT

TRAN- SACTION/ CART

Cart Transaction Design





- Home
- Product
- Cart
- Feedback

 Search



Jungkook



Feedback Form

If you have any questions or need more information about Superindo, or if you would like to provide suggestions, you can contact us by filling out the feedback form below:

Type of feedback

Complain



Name

Jeon Jungkook

Email

Jungkook.jeon@gmail.com

Write your Review

Write here...

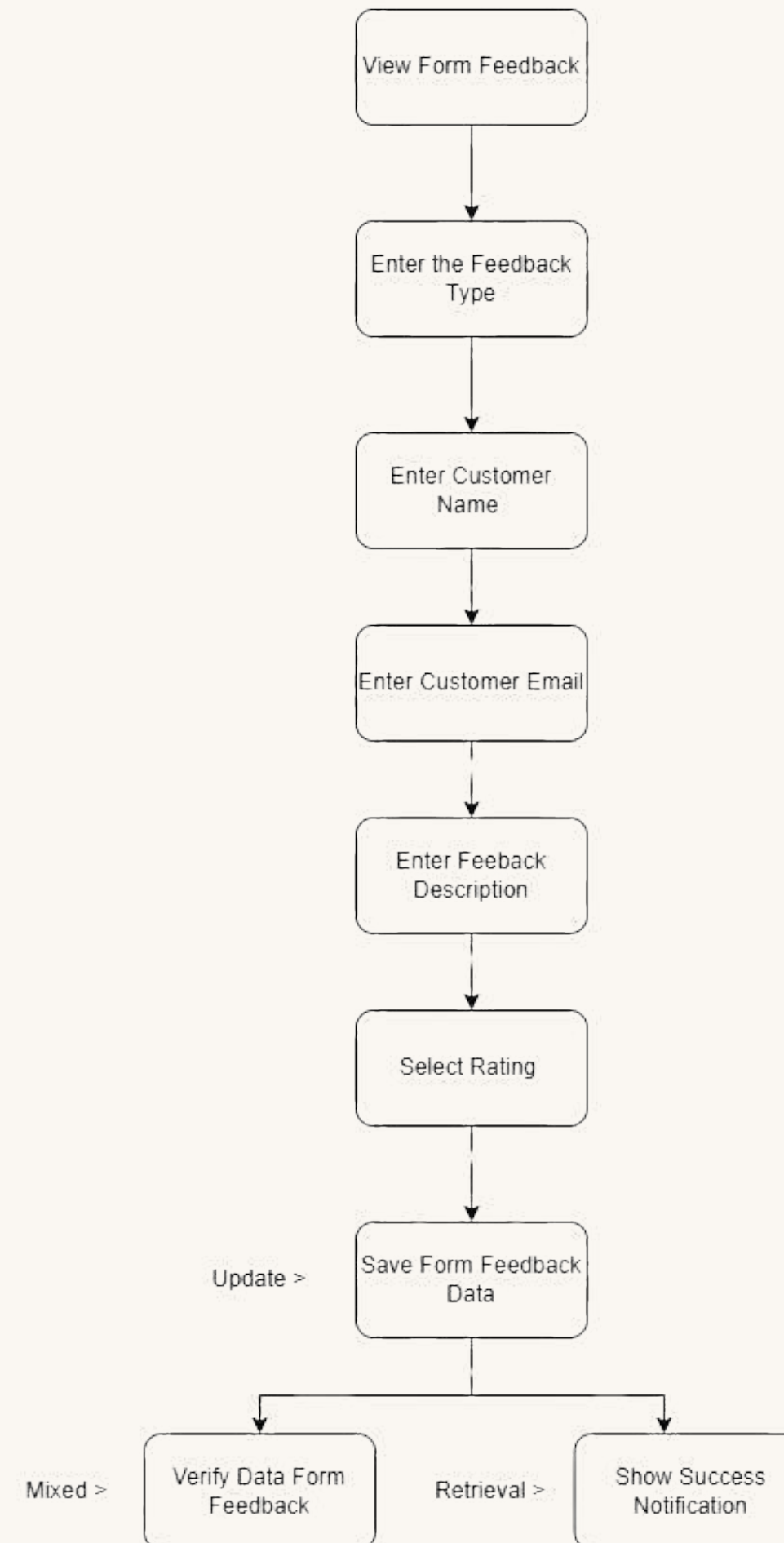
Add your Rating



Send

FEED BACK

Feedback Transaction Design



Data and Information Management

Table Instance Chart

Group 2

Table Instance Chart & Attribute Domains

Table Name : Customer

ColumnName	Customer ID	CustomerName	Customer Gender	CustomerBOD	Customer Address	Customer Email	Customer Phone
Key Type	PK						
Nulls/Unique	*	*	*	*	*	0	*
Sample Data	CU001	Anna Elsa	F	12/12/1999	Jl. Dahlia 2, No. 22, Jakarta Barat	Anna.elsa@gmail.com	0819227280
	CU002	Jason Crus	M	9/12/2000	Jl. Mawar, No. 80, Surabaya	Jason.crus@gmail.com	0880281923
	CU003	Jay Park	M	5/62002	Jl. Melati, No. 88, Surabaya	Jay.park@gmail.com	0839841098

*not null ; 0 null

Table Instance Chart & Attribute Domains

Table Name : Staff

Column Name	Staff ID	Staff Name	Staff Gender	StaffAddress	StaffBOD	Position	Staff Phone	Staff Email
Key Type	PK							
Nulls/Unique	*	*	*	*	*	*	*	o
Sample Data	ST001	Alman Cruz	M	Jl. Mawar, No. 200, Bekasi	6/7/1998	Cashier	0883920378	Alman.Criz@gmail.com
	ST002	Rizzy Tan	M	Jl. Belimbing, No. 99, Surabaya	19/4/2000	Store Manager	0829038616	Rizzy.Tan@gmail.com

Table Name : Transaction

ColumnName	TransactionID	CustomerID	ProductID	Transaction Date	Total Spending	Payment Method
Key Type	pk	fk	fk			
Nulls/Unique	*	*	*	*	*	*
Sample Data	TR001	CU003	PU001	06/1/2025	32.000	Cash
	TR002	CU001	PU001	06/2/2025	20.000	Debit

*not null ; o null

Table Instance Chart & Attribute Domains

Table Name : Membership

Column Name	Membership ID	CustomerID	Membershi pLevel	JoinDate	PointEarned	ExpiredDate	Status
Key Type	pk	fk					
Nulls/Uniqu e	*	*	*	*	o	*	*
Sample Data	MB001	CU001	Silver	6/2/2024	200	6/2/2030	Active
	MB002	CU002	Gold	2/1/2025	1000	2/1/2030	Active

Table Name : Products

Column Name	ProductID	ProductName	Category	Price	StockAvailability
Key Type	PK		FK		
Nulls/Unique	*	*	*	*	*
Sample Data	PU001	Indomie	10	3.500	500
	PU002	Ultra Milk	11	25.000	100
	PU003	Gula Pasir	12	13.000	250

*not null
o null

Table Instance Chart & Attribute Domains

Table Name : Feedback

Column Name	FeedbackID	CustomerID	FeedbackType	FeedbackDescription	FeedbackStatus
Key Type	PK	FK			
Nulls/Unique	*	*	*	*	*
Sample Data	FB001	CU003	Complaint	Harga di rak berbeda dengan harga di kasir	Pending
	FB002	CU002	Suggestion	Sebaiknya ada lebih banyak promo diskon	In Progress
	FB003	CU001	Praise	Pelayanan kasir sangat ramah dan cepat	Solved

Table Name : Regular

Column Name	CustomerID	Promo_Eligibility
Key Type	PK	
Nulls/Unique	*	*
Sample Data	CU005	YES
	CU007	YES

*not null
o null

Table Instance Revision

Table : Customer

ColumnName	Customer ID	Customer Name	Customer Gender	Customer BOD	Customer Address	Customer Email	Customer Phone
Key Type	PK						
Nulls/Unique	*	*	*	*	*	o	*
Sample Data	CU001	Anna Elsa	F	12/12/1999	Jl. Dahlia 2, No. 22, Jakarta Barat	Anna.elsa@gmail.com	0819227280
	CU002	Jason Crus	M	9/12/2000	Jl. Mawar, No. 80, Surabaya	Jason.crus@gmail.com	0880281923
	CU003	Jay Park	M	5/62002	Jl. Melati, No. 88, Surabaya	Jay.park@gmail.com	0839841098

*not null ; o null

Table Instance Revision

Table : Staff

ColumnName	Staff ID	Staff Name	Staff Gender	StaffAddress	StaffBOD	Position	Staff Phone	Staff Email
Key Type	PK							
Nulls/Unique	*	*	*	*	*	*	*	o
Sample Data	ST001	Alman Cruz	M	Jl. Mawar, No. 200, Bekasi	6/7/1998	Cashier	0883920378	Alman.Criz@gmail.com
	ST002	Rizzy Tan	M	Jl. Belimbing, No. 99, Surabaya	19/4/2000	Store Manager	0829038616	Rizzy.Tan@gmail.com

*not null ; o null

Table Instance Revision

Table : Transaction

ColumnName	TransactionID	CustomerID	Transaction Date	Total Spending	Payment Method
Key Type	pk	fk			
Nulls/Unique	*	*	*	*	*
Sample Data	TR001	CU003	06/1/2025	32.000	Cash
	TR002	CU001	06/2/2025	20.000	Debit

*not null ; o null

Table Instance Revision

Table : Transaction Detail

ColumnName	DetailID	TransactionID	ProductID	Quantity	Price	Discount	Subtotal
Key Type	pk	fk	fk				
Nulls/Unique	*	*	*	*	*	*	*
Sample Data	DT001	TR001	PU001	2	3.500	-	7.000
	DT002	TR001	PU002	1	25.000	-	25.000

*not null ; o null

Table Instance Revision

Table : Membership

Column Name	Members hipID	Customer ID	Membersh ipLevel	JoinDate	MemberPoint	ExpirationD ate	Status
Key Type	PK	PK					
Nulls/Unique	*	*	*	*	o	*	*
Sample Data	MB001	CU001	Silver	6/2/2024	200	6/2/2030	Active
	MB002	CU002	Gold	2/1/2025	1000	2/1/2030	Active

*not null ; o null

Table Instance Revision

Table : Product

Column Name	ProductID	ProductName	CategoryID	Price	StockAvailability
Key Type	PK		FK		
Nulls/Unique	*	*	*	*	*
Sample Data	PU001	Indomie	10	3.500	500
	PU002	Ultra Milk	11	25.000	100

*not null ; o null

Table Instance Revision

Table : Category

Column Name	CategoryID	Description
Key Type	pk	
Nulls/Unique	*	*
Sample Data	10	Makanan instan
	11	Minuman
	12	Bahan pokok

*not null ; o null

Table Instance Revision

Table : Feedback

Column Name	FeedbackID	CustomerID	FeedbackType	FeedbackDescription	FeedbackStatus
Key Type	PK	FK			
Nulls/Unique	*	*	*	*	*
Sample Data	FB001	CU003	Complaint	Harga di rak berbeda dengan harga di kasir	Pending
	FB002	CU002	Suggestion	Sebaiknya ada lebih banyak promo diskon	In Progress
	FB003	CU001	Praise	Pelayanan kasir sangat ramah dan cepat	Solved

*not null ; o null

Table Instance Revision

Table : Regular

Column Name	CustomerID	Promo_Eligibility
Key Type	PK	
Nulls/Unique	*	*
Sample Data	CU005	YES
	CU007	YES

*not null ; o null

Data and Information Management

Non-Relational Database

Group 2

Sample Data Relationship Database

Table: CUSTOMER

CustomerID	CustomerName	Gender	CustomerDOB	CustomerAddress	CustomerEmail	Phone
CU001	Anna Elsa	F	12/12/1999	Jl. Dahlia 2, No. 22, Jakarta Barat	Anna.elsa@gmail.com	0819227280
CU002	Jason Crus	M	09/1/2000	Jl. Mawar, No. 80, Surabaya	Jason.crus@gmail.com	0880281923
CU003	Jay Park	M	05/6/2002	Jl. Melati No. 88, Surabaya	Jay.park@gmail.com	0839841098

Table: TRANSACTION

TransactionID	StaffID	CustomerID	ProductID	Transaction Date	TotalSpending	PaymentMethod
TR001	ST001	CU003	PU001	6/1/2025	32.000	Cash
TR002	ST002	CU001	PU001	6/2/2025	20.000	Debit 

(Foreign key: CustomerID → Customer; StaffID → Staff; ProductID → Product)

Sample Data Relationship Database

Table: MEMBERSHIP

MembershipID	CustomerID	MembershipLevel	JoinDate	PointEarned	ExpiredDate	Status
MB001	CU001	Silver	6/2/2025	500	6/Feb/2030	Active
MB002	CU002	Gold	6/1/2025	10000	6/Jan/2030	Active

(Foreign key: CustomerID → Customer)

Table: STAFF

Staff ID	Staff Name	Staff Gender	Staff Address	Staff BOD	Position	Staff Phone	Staff Email
ST001	Alman Cruz	M	Jl. Mawar, No. 200, Bekasi	6/7/1998	Cashier	0883920378	Alman.Criz@gmail.com
ST002	Rizzy Tan	M	Jl. Belimbing, No. 99, Surabaya	19/4/2000	Store Manager	0829038616	Rizzy.Tan@gmail.com

Sample Data Relationship Database

Table: PRODUCT

ProductID	Product_Name	Category	Price	Stock_Availability
PU001	Indomie	10	3.500	500
PU002	Ultra Milk	11	25.000	100
PU003	Gula Pasir	12	13.000	250

Table: FEEDBACK

FeedbackID	CustomerID	Feedback_Type	Feedback_Description	Feedback_Status
FB001	CU003	Complaint	Harga di rak berbeda dengan harga di kasir	Pending
FB002	CU002	Suggestion	Sebaiknya ada lebih banyak promo diskon	In Progress
FB003	CU001	Praise	Pelayanan kasir sangat ramah dan cepat	Solved

(Foreign key: CustomerID → Customer)

Table: REGULAR

CustomerID	Promo_Eligibility
CU005	YES 
CU007	YES

(Foreign key: CustomerID → Customer) 

Sample of Document Database

Tabel Customer

```
{_id: CU001,  
  CustomerName: "Anna Elsa",  
  CustomerGender: "F",  
  CustomerBOD: 12/12/1999,  
  CustomerAddress: "Jl. Dahlia 2, No. 22, Jakarta Barat",  
  CustomerEmail: "Anna.elsa@gmail.com",  
  CustomerPhone : "0819227280"  
}
```

```
{_id: CU002,  
  CustomerName: "Jason Crus",  
  CustomerGender: "M",  
  CustomerBOD: 9/1/2000,  
  CustomerAddress: "Jl. Mawar, No. 80, Surabaya",  
  CustomerEmail: "Jason.crus@gmail.com",  
  CustomerPhone : "0880281923"  
}
```

```
{_id: CU003,  
  CustomerName: "Jay Park",  
  CustomerGender: "M",  
  CustomerBOD: 5/6/2002,  
  CustomerAddress: "Jl. Melati, No. 88, Surabaya",  
  CustomerEmail: "Jay.park@gmail.com",  
  CustomerPhone : "0839841098"  
}
```


Tabel Staff

```
{_id: ST001,  
StaffName: "Alman Cruz",  
StaffGender: "M",  
StaffBOD: 6/07/1998,  
StaffAddress: "Jl. Mawar, No. 200, Bekasi",  
StaffEmail: "Alman.cruz@gmail.com",  
StaffPhone : "083920378"  
Position : "Cashier"  
}  
  
{_id: ST002,  
StaffName: "Rizzy Tan",  
StaffGender: "M",  
StaffBOD: 19/4/2000,  
StaffAddress: "Jl. Belimbing, No. 99, Surabaya",  
StaffEmail: "Rizzy.tan@gmail.com",  
StaffPhone : "0829038616"  
Position : "Store Manager"  
}
```

Tabel Transaction

```
{_id: TR001,  
CustomerID: CU003,  
TransactionDate: 6/01/2025,  
TotalSpending: 32.000,  
PaymentMethod: "Cash"  
}  
  
{_id: TR002,  
CustomerID: CU001,  
TransactionDate: 6/02/2025,  
TotalSpending: 20.000,  
PaymentMethod: "Debit"  
}
```

Tabel Regular

```
{_id: CU005,  
PromoEligibility: "YES"  
}  
  
{_id: CU007,  
PromoEligibility: "YES"  
}
```

Tabel Membership

```
{_id: MB001,  
CustomerID: "CU001",  
MembershipLevel: "Silver",  
JoinDate: 6/02/2025,  
PointEarned: 200,  
ExpiredDate: 6/02/2030,  
Status: "Active"  
}
```

```
{_id: MB002,  
CustomerID: "CU002",  
MembershipLevel: "Gold",  
JoinDate: 2/01/2025,  
PointEarned: 1000,  
ExpiredDate: 2/01/2030,  
Status: "Active"  
}
```

Tabel Product

```
{_id: PU001,  
ProductName: "Indomie",  
Category: 10,  
Price: 3.500,  
StockAvailability: 500  
}
```

```
{_id: PU002,  
ProductName: "Ultra Milk",  
Category: 11,  
Price: 25.000,  
StockAvailability: 100  
}
```

```
{_id: PU003,  
ProductName: "Indomie",  
Category: 12,  
Price: 13.000,  
StockAvailability: 250  
}
```

Tabel Feedback

```
{_id: FB001,  
CustomerID: "CU001",  
FeedbackType: "Complaint",  
FeedbackDescription: "Harga di rak  
berbeda dengan harga di kasir",  
FeedbackStatus: "Pending"  
}
```

```
{_id: FB002,  
CustomerID: "CU002",  
FeedbackType: "Suggestion",  
FeedbackDescription: "Sebaiknya ada  
lebih banyak promo diskon",  
FeedbackStatus: "In Progress"  
}
```

```
{_id: FB003,  
CustomerID: "CU003",  
FeedbackType: "Praise",  
FeedbackDescription: "Pelayanan kasir  
sangat ramah dan cepat",  
FeedbackStatus: "Solved"  
}
```

Sample of Key-Value Database

Table Customer

Customer:CU001:CustomerName = "Anna Elsa"
Customer:CU001:CustomerGender = "F"
Customer:CU001:CustomerBOD = 12/12/1999
Customer:CU001:CustomerAddress = "Jl. Dahlia 2, No. 22, Jakarta Barat"
Customer:CU001:CustomerEmail = "Anna.elsa@gmail.com"
Customer:CU001:CustomerPhone = "019227280"

Customer:CU002:CustomerName = "Jason Crus"
Customer:CU002:CustomerGender = "M"
Customer:CU002:CustomerBOD = 9/1/2000
Customer:CU002:CustomerAddress = "Jl. Mawar, No. 80, Surabaya"
Customer:CU002:CustomerEmail = "Jason.crus@gmail.com"
Customer:CU002:CustomerPhone = "080281923"

Customer:CU003:CustomerName = "Jay Park"
Customer:CU003:CustomerGender = "M"
Customer:CU003:CustomerBOD = 5/6/2002
Customer:CU003:CustomerAddress = "Jl. Melati, No. 88, Surabaya"
Customer:CU003:CustomerEmail = "Jay.park@gmail.com"
Customer:CU003:CustomerPhone = "039841098"

Table Staff

Staff:ST001:StaffName = "Alman Cruz"
Staff:ST001:StaffGender = "M"
Staff:ST001:StaffAddress = "Jl. Mawar, No. 200, Bekasi"
Staff:ST001:StaffBOD = 6/7/1998
Staff:ST001:Position = "Cashier"
Staff:ST001:StaffPhone = "083920378"
Staff:ST001:StaffEmail = "Alman.Cruz@gmail.com"

Staff:ST002:StaffName = "Rizzy Tan"
Staff:ST002:StaffGender = "M"
Staff:ST002:StaffAddress = "Jl. Belimbing, No. 99, Surabaya "
Staff:ST002:StaffBOD = 19/4/2000
Staff:ST002:Position = "Store Manager"
Staff:ST002:StaffPhone = "029038616"
Staff:ST002:StaffEmail = "Rizzy.Tan@gmail.com"

Table Transaction

Transaction:TR001:TransactionDate = 6/1/2025
Transaction:TR001:TotalSpending = 32.000
Transaction:TR001:PaymentMethod = "Cash"

Transaction:TR002:TransactionDate = 6/2/2025
Transaction:TR002:TotalSpending = 20.000
Transaction:TR002:PaymentMethod = "Debit"

Table Product

Product:PU001:ProductName = "Indomie"
Product:PU001:Category = 10
Product:PU001:Price = 3.500
Product:PU001:Stock_Availability = 500

Product:PU002:ProductName = "Ultra Milk"
Product:PU002:Category = 11
Product:PU002:Price = 25.000
Product:PU002:Stock_Availability = 100

Product:PU003:ProductName = "Gula Pasir"
Product:PU003:Category = 12
Product:PU003:Price = 13.000
Product:PU003:Stock_Availability = 250

Table Membership

Membership:MB001:MembershipLevel = "Silver"
Membership:MB001:JoinDate = 6/2/2025
Membership:MB001:PointEarned = 200
Membership:MB001:ExpiredDate = 6/2/2030
Membership:MB001:Status = "Active"

Membership:MB002:MembershipLevel = "Gold"
Membership:MB002:JoinDate = 2/1/2025
Membership:MB002:PointEarned = 1000
Membership:MB002:ExpiredDate = 2/1/2030
Membership:MB002:Status = "Active"

Table Feedback

Feedback:FB001:Feedback_Type = "Complaint"
Feedback:FB001:Feedback_Description = "Harga di rak berbeda dengan harga di kasir"
Feedback:FB001:Feedback_Status = "Pending"

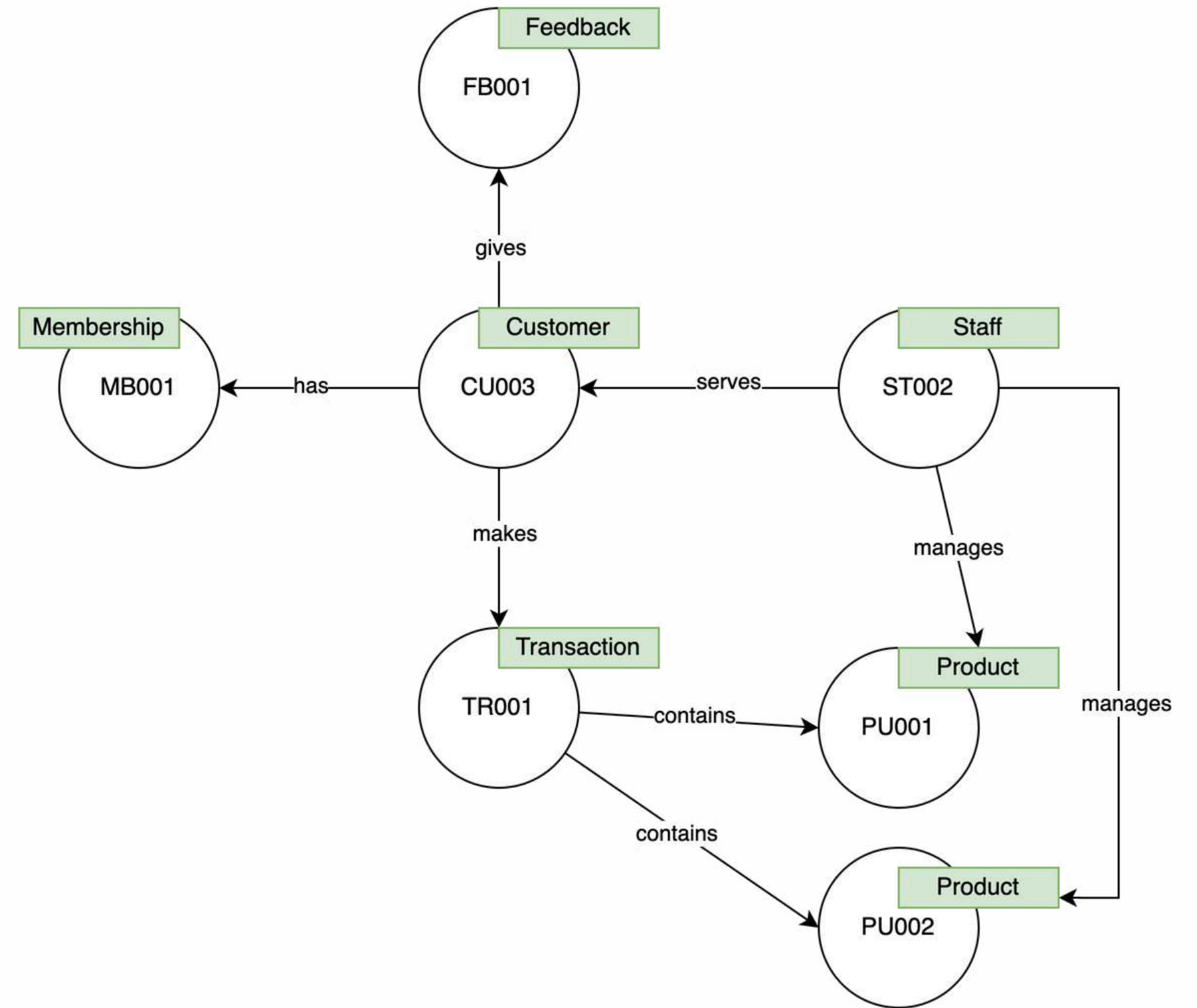
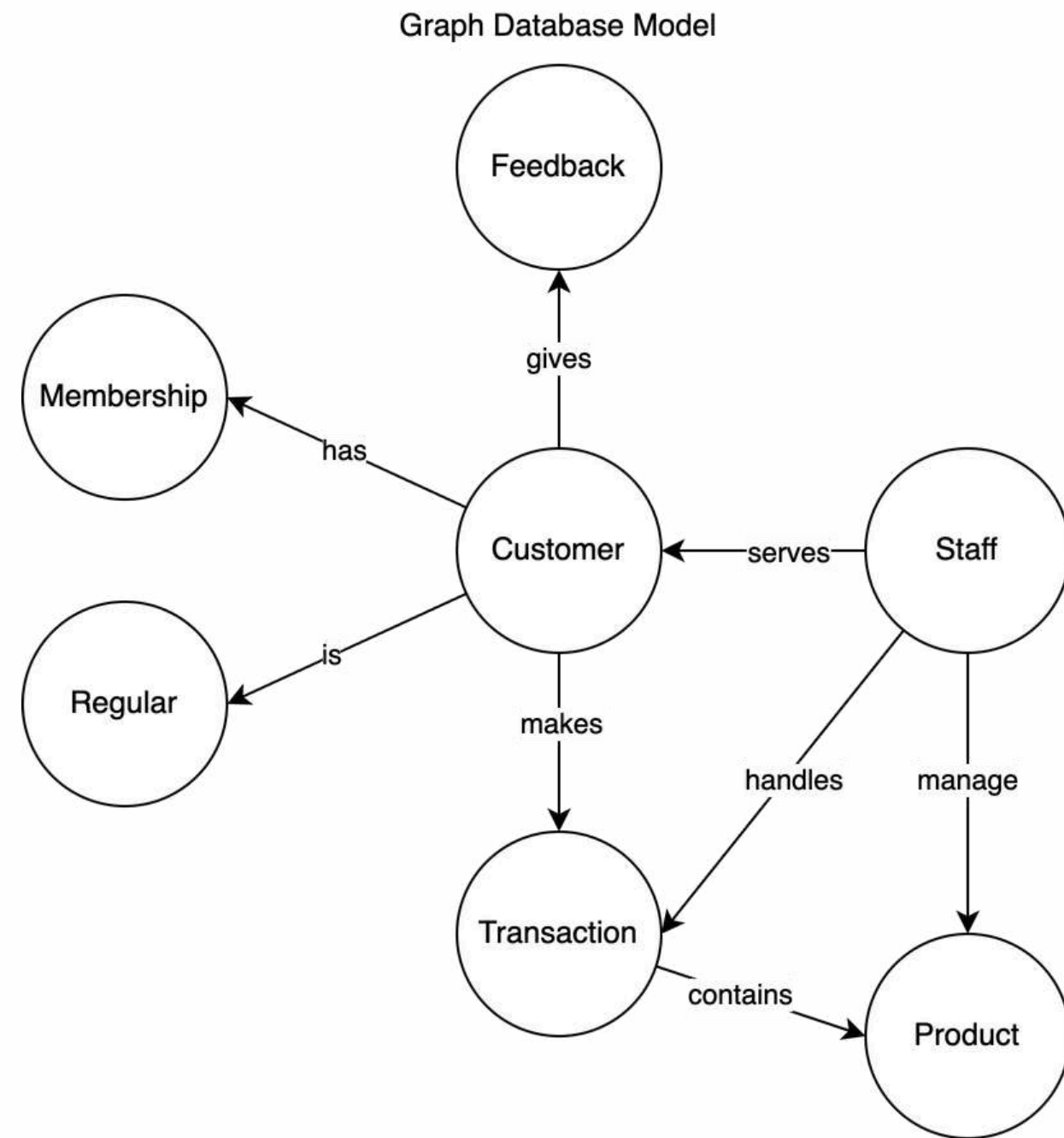
Feedback:FB002:Feedback_Type = "Suggestion"
Feedback:FB002:Feedback_Description = "Sebaiknya ada lebih banyak promo diskon"
Feedback:FB002:Feedback_Status = "In Progress"

Feedback:FB003:Feedback_Type = "Praise"
Feedback:FB003:Feedback_Description = "Pelayanan kasir sangat ramah dan cepat"
Feedback:FB003:Feedback_Status = "Solved"

Table Regular

Regular:CU005:Promo_Eligibility = "YES"
Regular:CU007:Promo_Eligibility = "YES"

Sample of Graph Database



Sample of Column Family Database



CU001	CustomerName	CustomerGender	CustomerBOD	CustomerAddress	CustomerEmail	CustomerPhone
	Anna Elsa	F	12/12/1999	Jl. Dahlia 2, No. 22, Jakarta Barat	Anna.elsa@gmail.com	0819227280
	1717091200000	1717091200000	1717091200000	1717091200000	1717091200000	1717091200000

CU002	CustomerName	CustomerGender	CustomerBOD	CustomerAddress	CustomerEmail	CustomerPhone
	Jason Crus	M	9/1/2000	Jl. Mawar No.80, Surabaya	Jason.crus@gmail.com	0880281923
	1717091210000	1717091210000	1717091210000	1717091210000	1717091210000	1717091210000

CU003	CustomerName	CustomerGender	CustomerBOD	CustomerAddress	CustomerEmail	CustomerPhone
	Jay Park	M	5/6/2002	Jl. Melati, No. 88, Surabaya	Jay.park@gmail.com	0839841098
	1717091220000	1717091220000	1717091220000	1717091220000	1717091220000	1717091220000

Sample of Column Family Database

Staff

ST001

StaffName

Alman Cruz

1718091200000

StaffGender

M

1718091200000

StaffAddress

Jl. Mawar, No. 200, Bekasi

1718091200000

StaffBOD

6/7/1998

1718091200000

StaffPosition

Cashier

1718091200000

StaffEmail

Alman.cruz@gmail.com

1718091200000

StaffPhone

083920378

1718091200000

ST002

StaffName

Rizzy Tan

1718091300000

StaffGender

M

1718091300000

StaffAddress

Jl. Belimbing, No. 99, Surabaya

1718091300000

StaffBOD

19/4/2000

1718091300000

StaffPosition

Store Manager

1718091300000

StaffEmail

Rizzy.tan@gmail.com

1718091300000

StaffPhone

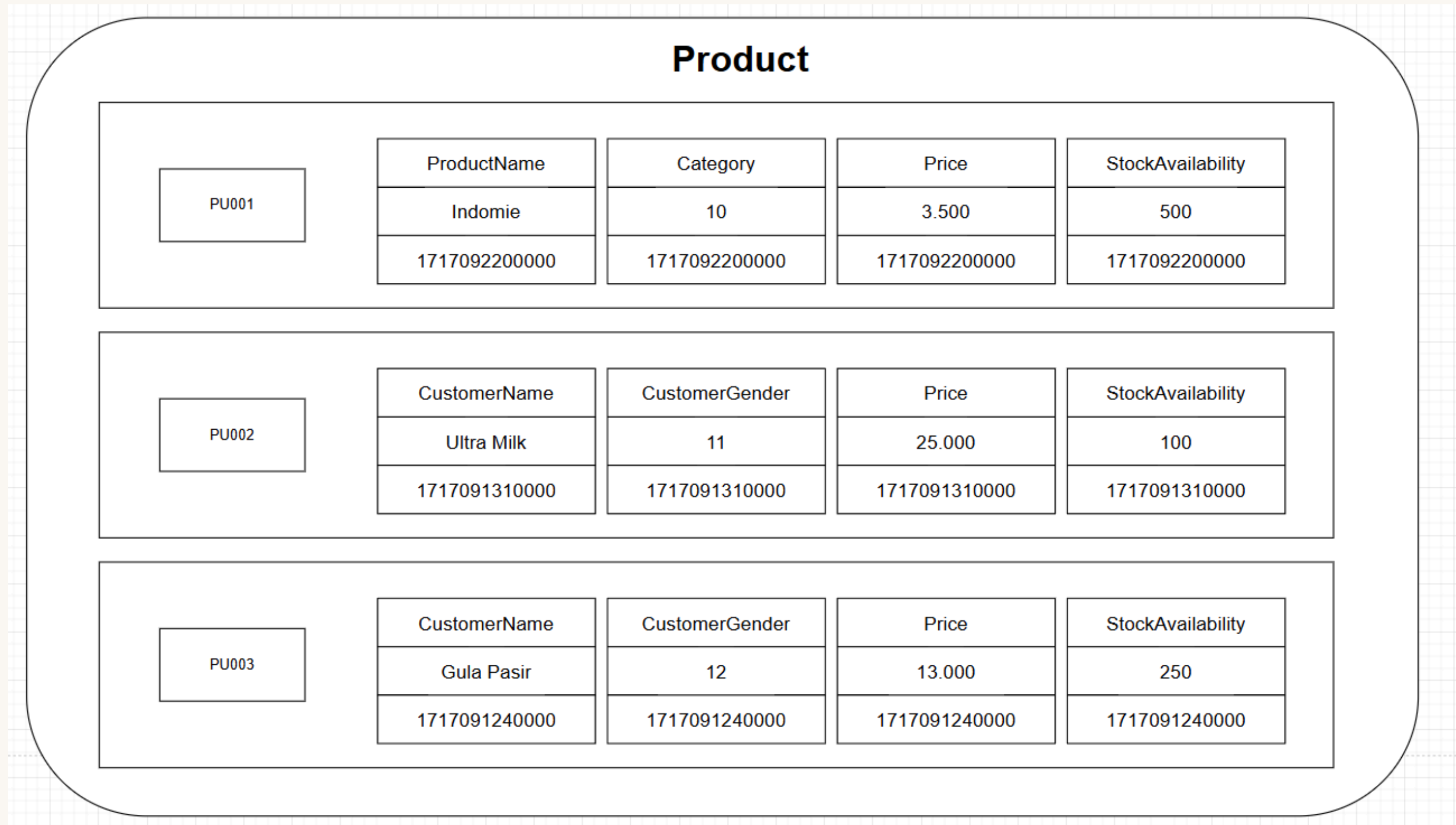
0829038616

1718091300000

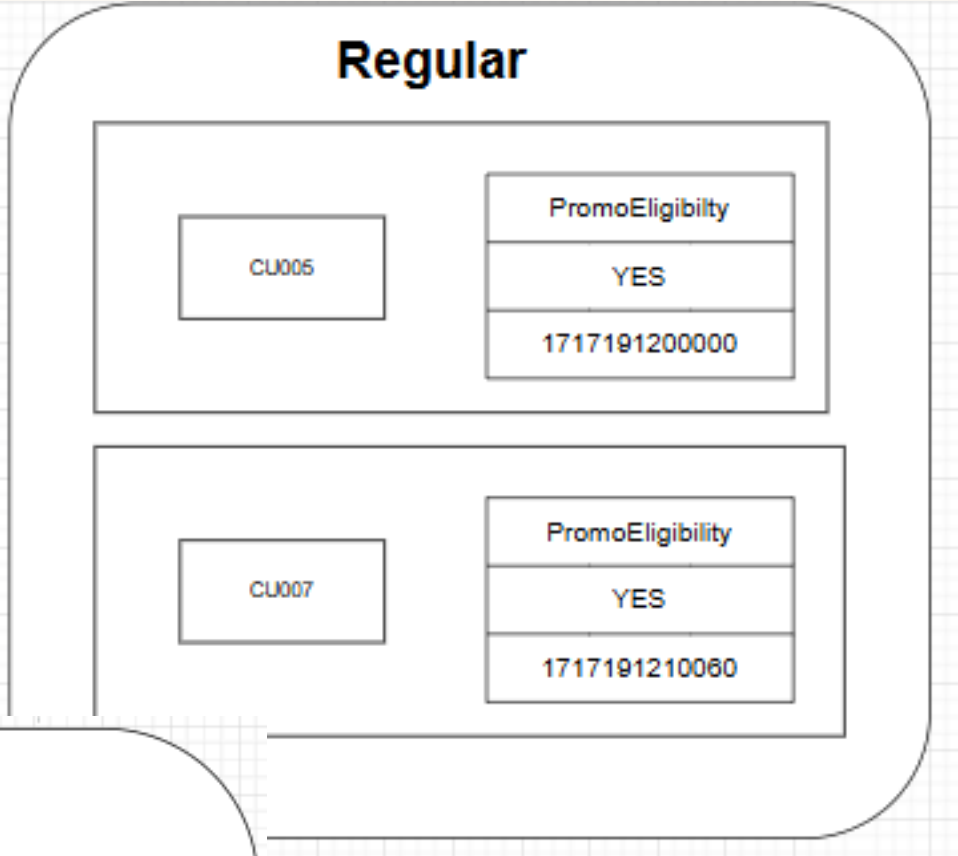
Sample of Column Family Database

Transaction						
TR001	StaffID	CustomerID	ProductID	TransactionDate	TotalSpending	PaymentMethod
	ST001	CU003	PU001	6/1/2024	32.000	Cash
	1717091250000	1717091250000	1717091250000	1717091250000	1717091250000	1717091250000
TR002	StaffID	CustomerID	ProductID	TransactionDate	TotalSpending	PaymentMethod
	ST002	CU002	PU001	2/1/2025	20.000	Debit
	1717091255000	1717091255000	1717091255000	1717091255000	1717091255000	1717091255000

Sample of Column Family Database



Sample of Column Family Database



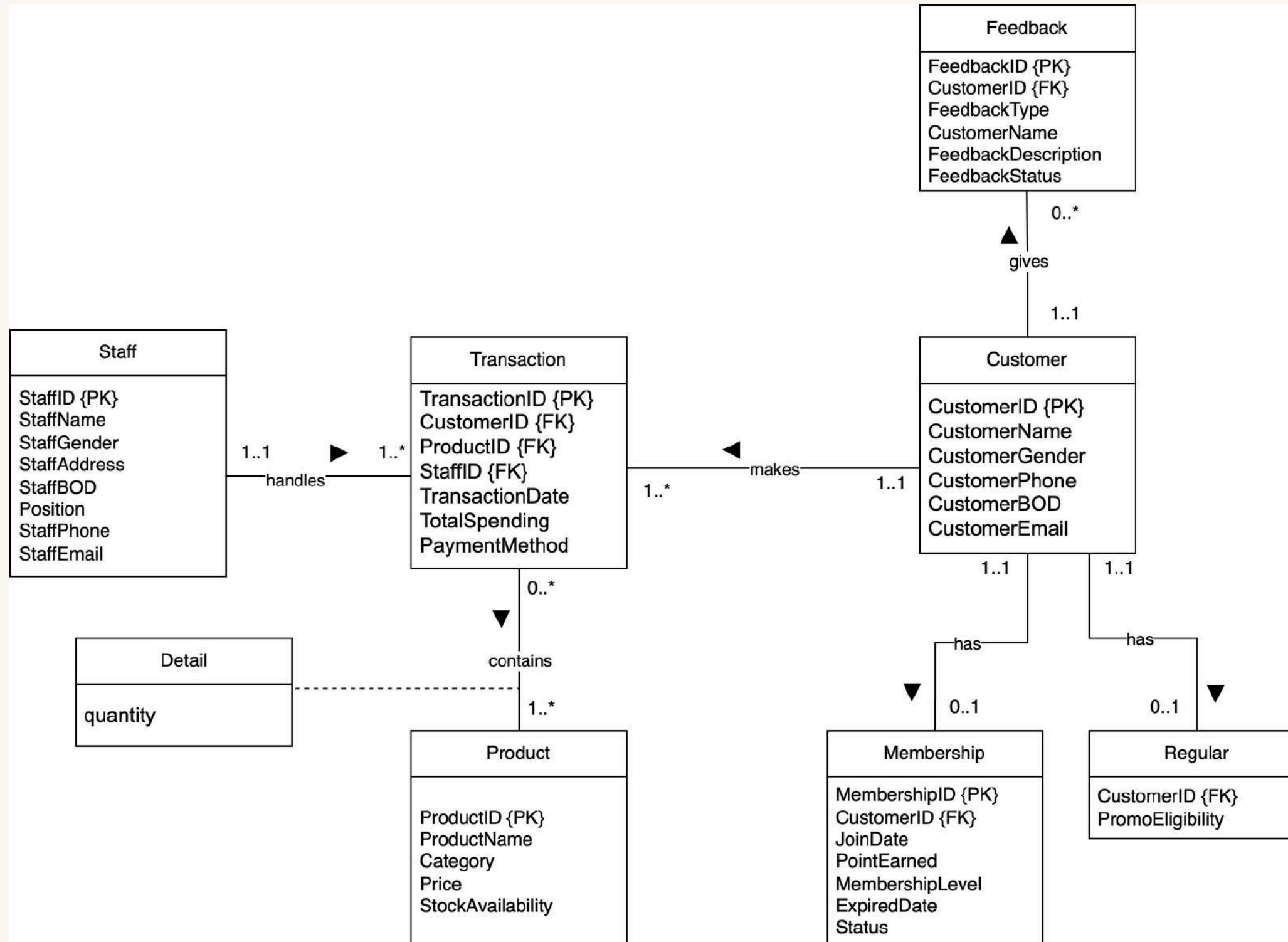
Membership						
MB001	CustomerID	MembershipLevel	JoinDate	PointEarned	ExpiredDate	Status
	CU003	Silver	6/2/2025	200	6/2/2030	Active
	1717091250000	1717091250000	1717091250000	1717091250000	1717091250000	17170912500000
MB002	CustomerName	MembershipLevel	JoinDate	PointEarned	ExpiredDate	Status
	CU002	Gold	2/1/2025	1000	2/1/2030	Active
	1717091240000	1717091240000	1717091240000	1717091240000	1717091240000	1717091240000

Sample of Column Family Database

Feedback				
FB001	CustomerID	FeedbackType	FeedbackDescription	FeedbackStatus
	CU003	Complaint	Harga di rak berbeda dengan harga di kasir	Pending
	1717092200000	1717092200000	1717092200000	1717092200000
FB002	CustomerID	FeedbackType	FeedbackDescription	FeedbackStatus
	Ultra Milk	Suggestion	Sebaiknya ada lebih banyak promo diskon	In Progress
	1717091310000	1717091310000	1717091310000	1717091310000
FB003	CustomerID	FeedbackType	FeedbackDescription	FeedbackStatus
	Gula Pasir	Praise	Pelayanan kasir sangat ramah dan cepat	Solved
	1717091240000	1717091240000	1717091240000	1717091240000

Consider Controlled Redudancy

Sample Relation Diagram



Sample Relations

Customer

Customer ID	Customer Name	Customer Gender	Customer BOD	Customer Address	Customer Email	Customer Phone
CU001	Anna Elsa	F	12/12/1999	Jl. Dahlia 2, No. 22, Jakarta Barat	Anna.elsa@gmail.com	0819227280
CU002	Jason Crus	M	09/01/2000	Jl. Mawar, No. 80, Surabaya	Jason.crus@gmail.com	0880281923
CU003	Jay Park	M	05/06/2002	Jl. Melati, No. 88, Surabaya	Jay.park@gmail.com	0839841098

Staff

Staff ID	Staff Name	Staff Gender	StaffAddress	StaffBOD	Position	Staff Phone	Staff Email
ST001	Alman Cruz	M	Jl. Mawar, No. 200, Bekasi	06/07/1998	Cashier	0883920378	Alman.Criz@gmail.com
ST002	Rizzy Tan	M	Jl. Belimbing, No. 99, Surabaya	19/04/2000	Store Manager	0829038616	Rizzy.Tan@gmail.com

Sample Relations

Transaction

TransactionID	StaffID	CustomerID	ProductID	Transaction Date	Total Spending	Payment Method
TR001	ST001	CU003	PU001	06/01/2025	32.000	Cash
TR002	ST001	CU001	PU001	06/02/2025	20.000	Debit

Membership

MembershipID	CustomerID	Membership Level	JoinDate	MemberPoint	ExpiredDate	Status
MB001	CU003	Silver	06/02/2024	200	06/02/2030	Active
MB002	CU002	Gold	02/01/2025	1000	02/01/2030	Active

Sample Relations

Product

ProductID	ProductName	Category	Price	Stock_Availability
PU001	Indomie	10	3.500	500
PU002	Ultra Milk	11	25.000	100
PU003	Gula Pasir	12	13.000	250

Reguler

CustomerID	Promo_Eligibility
CU005	YES
CU007	YES

Feedback

FeedbackID	CustomerID	Feedback_Type	Feedback_Description	Feedback_Status
FB001	CU003	Complaint	Harga di rak berbeda dengan harga di kasir	Pending
FB002	CU002	Suggestion	Sebaiknya ada lebih banyak promo diskon	In Progress
FB003	CU001	Praise	Pelayanan kasir sangat ramah dan cepat	Solved

Sample Relations

Step 7.1 Combining 1:1 relationships

There is no 1:1 relationships

Step 7.2 Duplicating non-key attributes in 1:* relationships to reduce joins

Dari table Product

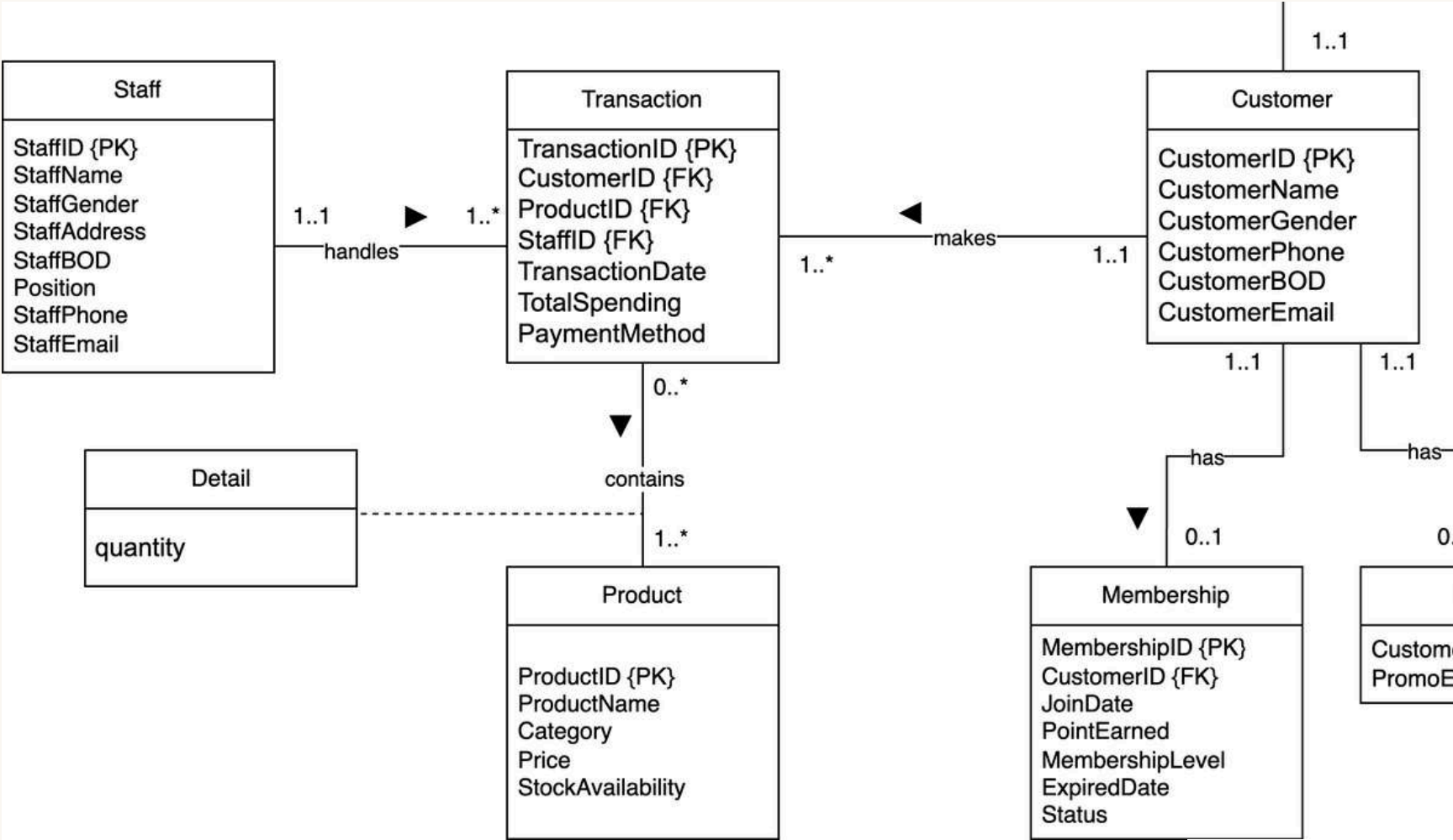
Category	Description
10	Makanan instan
11	Minuman
12	Bahan pokok

ProductID	ProductName	Category	Price	StockAvailability
PU001	Indomie	10	3.500	500
PU002	Ultra Milk	11	25.000	100
PU003	Gula Pasir	12	13.000	250

ProductID	ProductName	Category	Description	Price	StockAvailability
PU001	Indomie	10	Makanan Instan	3.500	500
PU002	Ultra Milk	11	Minuman	25.000	100
PU003	Gula Pasir	12	Bahan pokok	13.000	250

Sample Relations

Step 7.3 Duplicating FK attributes in 1:* relationships to reduce joins



TransactionID	StaffID	StaffName	CustomerID	CustomerName	ProductID	ProductName	TransactionDate	TotalSpending	PaymentMethod
TR001	ST001	Alman Cruz	CU003	Jay Park	PU001	Indomie	06/01/2025	32.000	Cash
TR002	ST001	Alman Cruz	CU001	Anna Elsa	PU001	Indomie	06/02/2025	20.000	Debit

Sample Relations

Step 7.4 Duplicating attributes in *:~ relationships to reduce joins

TransactionID	StaffID	StaffName	CustomerID	CustomerName	ProductID	ProductName	Transaction Date	Total Spending	Payment Method
TR001	ST001	Alman Cruz	CU003	Jay Park	PU001	Indomie	06/01/2025	32.000	Cash
TR002	ST001	Alman Cruz	CU001	Anna Elsa	PU001	Indomie	06/02/2025	20.000	Debit

Step 7.5 Introducing repeating groups

TransactionID	StaffID	Customer ID	ProductID	ProductID2	Transaction Date	Total Spending	Payment Method
TR003	ST001	CU003	PU001	PU002	06/01/2025	30.000	Cash

Sample Relations

Step 7.6 Creating extract tables

Laporan operasional seperti total penjualan harian, performa staf, atau produk terlaris memerlukan data hasil gabungan (join) dari beberapa relasi dasar seperti Transaction, Staff, Product, dan Customer.

Karena laporan tidak selalu memerlukan data real-time, maka memungkinkan untuk membuat tabel ekstrak terdenormalisasi yang dihasilkan dari join antar tabel dasar, agar proses akses data untuk pelaporan menjadi lebih cepat dan efisien.

Extract Table dapat mencakup data berikut:

TransactionID, TransactionDate
CustomerName, CustomerType(Subtypes)
StaffName

//ProductName, Quantity, Subtotal (TransactionDetail) ditanyakan/bingung maaf ya

TotalSpending, PaymentMethod

Dengan begitu, pengguna cukup akses Extract Table secara langsung tanpa perlu mengakses banyak tabel dasar setiap kali laporan dibutuhkan.

Sample Relations

Step 7.7 Partitioning relations

- **Horizontal Partitioning**

Penerapan:

Tabel Transaction dipartisi secara horizontal berdasarkan wilayah cabang atau rentang waktu (per bulan atau tahun). Misalnya, transaksi yang dilakukan di cabang Jakarta dan cabang Surabaya akan disimpan di partisi yang berbeda, atau transaksi tahun 2024 dan 2025 dipisahkan ke dalam partisi tahunan.

Keuntungan:

Mempercepat akses data transaksi pada waktu tertentu atau cabang tertentu.

Mengurangi jumlah data yang harus diproses dalam satu query.

Membantu dalam backup dan arsitektur sistem terdistribusi antar cabang.

- **Vertical Partitioning**

Penerapan:

Tabel Customer dipartisi secara vertikal menjadi dua bagian:

- Partisi pertama menyimpan atribut dasar yang sering diakses seperti CustomerID, CustomerName, CustomerPhone.
- Partisi kedua menyimpan atribut tambahan seperti CustomerAddress, CustomerBOD, CustomerEmail yang lebih jarang digunakan untuk operasional harian.

Tabel Product juga dapat dipartisi secara vertikal:

Atribut frequently-used seperti ProductID, ProductName, Price, StockAvailability disimpan terpisah dari atribut tambahan seperti Category.

Keuntungan:

- Mengurangi ukuran data yang dibaca sistem saat query rutin.
- Meningkatkan performa sistem karena hanya atribut penting yang diproses dalam query utama.
- Efisien untuk laporan harian seperti “produk dengan stok terendah” atau “pelanggan aktif”.

Data and Information Management

Big Data & AI Implementation

Group 2

Big Data

Big Data merujuk pada kumpulan data besar yang beragam, kompleks, dan terus berkembang—dan ini sangat relevan bagi Superindo yang mengelola lebih dari 200 gerai di berbagai kota. Setiap hari, sistem database Superindo menerima input ribuan hingga jutaan data transaksi, produk, dan feedback pelanggan. Melalui proses penyimpanan, pengolahan, dan analisis data yang sistematis.

Big Data memungkinkan Superindo untuk:

- Mengenal pelanggannya lebih dalam, misalnya dengan menganalisis kebiasaan belanja, waktu favorit berbelanja, hingga produk-produk yang paling sering dibeli.
- Mengantisipasi kebutuhan stok secara akurat, misalnya dengan mendeteksi kapan suatu produk mulai laris di wilayah tertentu, sehingga logistik bisa segera merespons.
- Meningkatkan efektivitas program loyalitas, dengan memberikan reward atau promo berdasarkan perilaku individual, bukan lagi satu pola untuk semua pelanggan.
- Menyatukan data dari berbagai sistem (kasir, online, keanggotaan, gudang) dalam satu platform terintegrasi yang memudahkan pelaporan dan pengambilan keputusan.

Artificial Intelligence

Setelah data terkumpul dengan baik, langkah selanjutnya adalah bagaimana memanfaatkannya untuk mengambil keputusan otomatis dan cerdas. Inilah peran AI dalam sistem Superindo.

Terdapat tiga pendekatan utama AI yang bisa diterapkan dalam konteks sistem yang telah dibangun:

1. Supervised Learning

Digunakan untuk memprediksi tingkat loyalitas pelanggan berdasarkan riwayat transaksi, level membership, dan jenis feedback yang mereka berikan. Dengan algoritma seperti Decision Tree atau SVM, Superindo dapat mengidentifikasi pelanggan yang berisiko berhenti belanja, lalu menawarkan insentif atau promo agar mereka kembali aktif.

2. Unsupervised Learning

Cocok untuk mengelompokkan pelanggan berdasarkan perilaku belanja, seperti kategori produk favorit atau metode pembayaran. Dengan clustering seperti K-Means, perusahaan bisa membuat kampanye iklan yang lebih personal dan menargetkan segmen pelanggan tertentu secara efisien.

3. Reinforcement Learning

Ideal untuk skenario pengelolaan stok dan penentuan harga dinamis. Sistem bisa belajar dari hasil penjualan sebelumnya dan menyesuaikan harga agar tetap kompetitif, terutama saat momen khusus seperti awal bulan atau hari libur.

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Implementasi Nyata dalam Sistem yang Dirancang

Sistem database yang telah dibangun dalam proyek ini telah disesuaikan untuk memfasilitasi integrasi Big Data dan AI. Tabel-tabel seperti Transaction, Membership, Customer, dan Feedback menyediakan data kunci yang dibutuhkan oleh algoritma pembelajaran mesin. Sistem juga dirancang dengan struktur yang fleksibel untuk memudahkan proses ETL (Extract, Transform, Load), serta partisi data yang mendukung kecepatan query dan pemrosesan data dalam jumlah besar.

Beberapa contoh penerapan konkret antara lain:

- Sistem rekomendasi produk otomatis di aplikasi Superindo berdasarkan riwayat belanja masing-masing pelanggan.
- Notifikasi otomatis saat pelanggan layak naik level membership, berdasarkan total transaksi dalam kurun waktu tertentu.
- Peringatan dini saat stok produk tertentu mulai menipis, berdasarkan tren penjualan mingguan atau musiman.

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Tantangan dan Solusi

Tentu saja, penerapan Big Data dan AI tidak lepas dari tantangan. Beberapa yang perlu diperhatikan oleh manajemen PT. Superindo antara lain:

- Kebutuhan akan infrastruktur IT yang memadai, seperti server, penyimpanan cloud, dan platform pemrosesan data real-time.
- Kualitas dan konsistensi data, yang harus dijaga agar hasil analisis dan prediksi tidak bias atau menyesatkan.
- Keamanan data pelanggan, khususnya karena sistem menyimpan informasi sensitif seperti email, nomor telepon, dan riwayat belanja.
- Kesiapan SDM, baik dari sisi tim IT maupun divisi operasional untuk bisa memanfaatkan insight yang diberikan oleh sistem AI.

Untuk menjawab tantangan tersebut, disarankan agar Superindo:

- Mengembangkan data governance yang kuat.
- Melatih staf dalam literasi data dan teknologi.
- Menerapkan enkripsi dan kontrol akses ketat dalam sistem database yang dibangun.

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Integrasi **Big Data dan AI** bukan hanya sebuah inovasi, tetapi langkah strategis bagi PT. Superindo dalam menghadapi masa depan industri ritel yang semakin digital dan kompetitif. Sistem database yang telah dirancang oleh tim kami memberikan pondasi yang kuat untuk hal tersebut. Dengan mengelola data secara terstruktur dan cerdas, serta memanfaatkannya untuk analisis dan otomatisasi, Superindo akan mampu memberikan layanan yang lebih cepat, lebih personal, dan lebih efektif kepada pelanggan. Dengan demikian, transformasi digital berbasis data bukan hanya soal teknologi, tapi tentang bagaimana Superindo meningkatkan hubungan dengan pelanggannya, mengoptimalkan proses bisnis, dan menciptakan nilai baru di tengah persaingan pasar yang dinamis.

Data and Information Management

**Thank You
So Much!**

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